

Pairing, Stripes, and Competing Order in Coupled Hubbard Ladders

Introduction, Background, and Current Results

Ladder MPS+MF project
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Abstract

Weakly coupled two-leg Hubbard ladders provide a setting in which the formation of singlet pairs can be studied separately from the collective order selected by coupling correlated subsystems. This draft develops the motivation for a multichannel matrix-product-state plus mean-field study of superconducting, charge, and magnetic order, with particular attention to the relationship between stripe-centered pairing proposals and the broader picture of intertwined orders. It also collects the project's current local evidence. The isolated repulsive ladder exhibits a finite-system spin gap and substantial pairing correlations, whereas the latest square-array trajectories at zero nearest-neighbor interaction develop strong stripe textures. At an attractive interaction, two contrasting initial states instead approach a common paired state. The unfrustrated-cubic grid reaches stripes at all nine tested coordinates. Finer square cuts retain distinct textures near the boundary, with a small change in energy slope along the interaction cut but no resolved first-order singularity. All four repulsive- V square starts lose pairing, including both intertwined seeds. Both trellis seeds become paired in a skew one-ladder cell but develop stripes in a rectangular two-ladder cell, where large alternating relaxation persists. Complete retrospective measurements show that these striped endpoints retain short-distance singlet-pair correlations concentrated near hole-rich magnetic walls, while longer-distance correlations are strongly suppressed. Rung-leg signs predominantly survive, but reference and fit-window sensitivity preclude a single asymptotic decay exponent. All four newer square two-ladder runs at the attractive paired coordinates pass fixed-point gates and reproduce the paired state from both tested seeds. Within the paired square subset, connected spin correlations at the stripe wavevector decrease with pairing along a hopping cut but increase along an interaction cut. These observations sharpen the distinction between local pairing tendencies and the order selected in an array. The numerical results remain finite-length, finite-bond-dimension evidence, with explicit limits on self-consistency, response completeness, and thermodynamic interpretation. Equal average densities on all ladders also exclude charge redistribution between ladders and the associated family of parallel stripes.

Draft scope. Sections 1–2 are draft manuscript prose. Section 3 is a dated collection of results and interpretation, including unresolved points that must remain visible during manuscript development. The evidence cutoff is the synchronized analyses through 23 September 2026, including reanalysis of the completed September 22 measurements; the literature discussion includes the 24 September review of ladder-cuprate families, magnetic and electronic parameter estimates, pressure, carrier partition, and structural limitations. The stripe-orientation and density-constraint discussion was updated on 23 September. Appendix A identifies the local sources. The correlation analyses use these completed measurements and simulations; no additional many-body solve was performed for manuscript preparation.

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1 Introduction

The relationship between superconductivity, antiferromagnetism, and spatial charge organization is a central question in doped Mott insulators. The Hubbard model provides a minimal language for studying how kinetic energy and local repulsion generate collective behavior, while its extensions allow the influence of nonlocal interactions and lattice structure to be examined systematically [1–4]. In this setting, strong singlet correlations need not lead directly to superconductivity. Charge and spin order can develop from the same interacting electrons, and the scales associated with pair formation and collective phase coherence can differ. A microscopic account of superconductivity must therefore address the competing ways in which a correlated electronic state can organize.

Stripe order makes this issue particularly concrete. Experiments in lanthanum-based cuprates established charge modulations accompanied by incommensurate magnetic order, with antiferromagnetic domains separated by hole-rich regions [5]. Subsequent measurements found closely related magnetic excitation spectra in stripe-ordered and superconducting materials, motivating theories in which stripe correlations participate in the pairing mechanism [6]. A complementary view regards stripes as a competing organization of a doped Mott insulator whose short-distance singlet physics can also support superconductivity without requiring a particular stripe arrangement [3]. The resulting question concerns both the origin of pairing and the consequences of charge organization for the coherence of pairs.

The distinction between these views is sharpened by separating the existence of stripe correlations from their necessity for superconductivity. In stripe-ordered $\text{La}_{1.875}\text{Ba}_{0.125}\text{CuO}_4$, transport and magnetization measurements reveal superconducting correlations within the planes despite strong suppression of three-dimensional phase coherence [7]. A spatially sign-changing superconducting order parameter can frustrate interplane Josephson coupling and offers a specific explanation of this separation [8]. Such observations permit cooperation at the level of local pairing and competition at the level of collective order. They motivate treating superconducting, charge, and spin degrees of freedom together, as emphasized in the framework of intertwined orders [9, 10].

Modern Hubbard-model calculations reinforce the dependence of this relationship on the microscopic Hamiltonian. Complementary numerical methods find stripe order with small energy differences between wavelengths in a studied regime of the nearest-neighbor square-lattice model [11]. Other calculations find no superconducting ground state in specified regimes of that minimal model [12], while additional hopping terms can support pairing together with spatial charge modulation [13, 14]. Deliberately imposed hopping inhomogeneity can also enhance superconducting correlations on cylinders [15]. These results do not give a universal sign to the influence of stripes. They instead identify a competition that must be resolved at the level of a specified model, geometry, and numerical accuracy.

Two-leg ladders provide useful correlated building blocks for this problem. Their magnetic spectrum differs qualitatively from that of a single chain, and doped repulsive ladders can support a spin gap, bound holes, and slowly decaying singlet-pair correlations [16–18]. The corresponding low-energy description can contain a gapped spin sector and a gapless total-charge mode [19, 20]. High-accuracy numerical studies show that the relative decay of pair and density correlations requires careful control of open boundaries, bond dimension, and fit windows [21, 22]. Thus a ladder can exhibit substantial pairing tendencies while retaining the fluctuations characteristic of an effectively one-dimensional system.

Cuprate ladder materials make the role of transverse geometry an experimental question. SrCu_2O_3 and $\text{Sr}_2\text{Cu}_3\text{O}_5$ provide insulating two- and three-leg benchmarks, while $\text{La}_4\text{Sr}_{10}\text{Cu}_{24}\text{O}_{41}$ resolves one- and two-triplon excitations in approximately undoped ladders with stronger leg

exchange and appreciable cyclic exchange [23, 24]. These magnetic benchmarks constrain a microscopic description without establishing superconductivity. Superconductivity was discovered under pressure in $\text{Sr}_{0.4}\text{Ca}_{13.6}\text{Cu}_{24}\text{O}_{41.84}$, a member of the $\text{Sr}_{14-x}\text{Ca}_x\text{Cu}_{24}\text{O}_{41}$ chain-ladder family [25]. The same family exhibits ladder charge order and pressure-dependent transport, providing a setting in which pairing, charge organization, and interladder coherence can be compared [26–28]. Its ladder sheets have displaced neighbors connected by zigzag paths: a trellis network rather than an array with only aligned transverse bonds. The chain subsystem also acts as a charge reservoir, so varying composition or pressure changes more than the coupling between otherwise fixed ladders. These distinctions motivate a controlled comparison of transverse connectivities before attempting a material-specific phase diagram. They also require separating a model scan from a material fit: the stronger-rung regime explored here differs from the stronger-leg estimates in several spectroscopic analyses [24, 29, 30].

Coupling such ladders introduces a second question: which order develops when the subsystems reinforce one another? Stripe-centered theories provide an explicit route from spin-gap formation to pairing and then to Josephson coherence [31]. A spatially modulated repulsive Hubbard construction demonstrates how nearly decoupled ladders can supply a pairing scale while weak interladder coupling establishes superconducting order [32]. Matrix-product-state plus mean-field methods provide a practical numerical realization of the broader coupled-subsystem strategy [33, 34]. Their multichannel extension shows why charge and superconducting solutions must be retained and compared within a common framework [35].

The interaction between neighboring densities supplies an additional tuning parameter. Calculations on extended Hubbard ladders and cylinders find that attraction can enhance pairing over selected parameter ranges, while strong attraction can also favor charge segregation [36, 37]. Recent cuprate-ladder spectroscopy, interpreted together with DMRG, provides experimental motivation for considering attraction beyond the simple onsite Hubbard interaction [30]. Its implications for a coupled array remain to be determined: stronger binding changes both the properties of the individual ladder and the effective interladder couplings. Pair mobility, susceptibility, and competing order must be considered along with binding.

The present project investigates these issues in arrays of repulsive two-leg extended Hubbard ladders, treating the ladder with an MPS ground-state method and the transverse environment with a multichannel mean-field construction. The principal regime is $U/t = 8$ and density $n = 15/16$ per physical site. Rung hopping, nearest-neighbor interaction, and transverse connectivity are varied while pairing, normal exchange, and spin-resolved density fields remain available. In particular, the trellis comparison tests whether zigzag coupling changes the balance between static stripes and pairing at the same intraladder parameters, using both one- and two-ladder spatial cells. The immediate objective is to establish reproducible ordered solutions and understand how their competition relates to the underlying ladder correlations. The current evidence suggests that intrinsic pairing tendencies can coexist with an array tendency toward static stripes, and that attraction can reverse the observed outcome at a representative square-array point. We examine this relationship from both sides: pair correlations within stripe textures, followed by charge and spin correlations within uniformly paired states. Establishing the extent and thermodynamic stability of these behaviors requires the numerical controls discussed below.

2 Physical background and connection to the model

2.1 Local singlets, pairing correlations, and coherence

Three observables answer different questions. A spin gap measures the energy cost of the lowest accessible spin excitation at fixed particle number. A pair-binding energy compares even- and odd-particle sectors. A pair correlation function measures how a pair created in one region correlates with a pair removed elsewhere. None of these quantities alone establishes a nonzero superfluid stiffness or a superconducting transition temperature. For example, an isolated number-conserving ladder has zero anomalous expectation value even when its pair correlations decay algebraically.

For a spin-singlet bond operator,

$$\Delta_{ab} = \frac{c_{a\uparrow}c_{b\downarrow} - c_{a\downarrow}c_{b\uparrow}}{\sqrt{2}}, \quad P_{ab,cd} = \langle \Delta_{ab}^\dagger \Delta_{cd} \rangle, \quad (1)$$

the relative signs between different bond orientations characterize the internal pair structure. Opposite signs on leg and rung bonds are commonly described as *d*-wave-like in a two-leg ladder. The reduced lattice symmetry and discrete transverse momentum should be kept in mind when connecting this terminology to the gap of a two-dimensional cuprate.

An order parameter with a nonzero center-of-mass wavevector is a different property. Schematically,

$$\langle \Delta_b(R) \rangle = f_b [\Psi_0 + \Psi_Q \cos(QR + \phi)], \quad (2)$$

separates the internal bond form factor f_b from the spatial structure of the condensate. An internal leg/rung sign change alone is not evidence of a pair-density wave. Nor does a boundary-induced modulation of a predominantly uniform pair field establish a distinct finite-momentum superconducting phase [10].

2.2 Two views of stripes and their modern interpretation

In a common-Mott-physics interpretation, singlet and magnetic correlations provide a shared microscopic starting point for several possible ordered states. Static stripes are then one possible competitor, and their presence need not be a prerequisite for pairing [3]. In a stripe-centered interpretation, spatial organization helps create the conditions for pairing. The spin-gap proximity mechanism, for example, couples a conducting stripe to a magnetically correlated environment; pair hopping transmits a spin gap to the mobile carriers, while Josephson coupling acts at a separate stage to establish coherence [31].

These interpretations share an emphasis on strong correlations and singlets. Their sharper disagreement concerns whether stripe organization is an essential intermediate step. The observations of planar superconductivity in stripe-ordered LBCO and the associated layer-decoupling proposal show how static stripes can coexist with pairing while suppressing a particular route to bulk coherence [7, 8]. Conversely, coexistence in a model or material does not establish that stripes caused its pairing.

The relation to the pseudogap remains unresolved. NMR measurements in Eu-LSCO find spin-stripe order extending practically to the pseudogap endpoint and identify competition with superconductivity, providing evidence for a close connection without determining its causal origin [38]. The spin gap of an isolated ladder, an electronic pseudogap in a cuprate, and the superconducting quasiparticle gap are different observables. A calculation of one should not be presented as a measurement or explanation of the others.

2.3 What magnetic spectroscopy establishes

Incommensurate magnetic peaks can be understood through antiphase spin domains. With lattice spacing set to one, a schematic stripe texture is

$$\langle S_{x,y}^z \rangle = m(-1)^{x+y} \cos(\kappa x), \quad q_c = 2\kappa, \quad \mathbf{q}_s = (\pi \pm \kappa, \pi). \quad (3)$$

The charge period is half the period of the staggered spin envelope. This spatial relation is useful evidence for an antiphase stripe texture, but its presence on a finite open ladder does not determine the thermodynamic amplitude or preferred wavelength.

The similarity between magnetic spectra in stripe-ordered LBCO and YBCO was an important motivation for stripe-centered interpretations [6]. Quantum stripe calculations show that low-energy spin-wave-like excitations and higher-energy ladder-like triplet excitations can occur within the same ordered model [39]. Consequently, a failure of a simple semiclassical intensity pattern does not rule out stripes or all low-energy spin waves. The connection between a spectral shape and a microscopic pairing mechanism requires more than qualitative resemblance.

The magnetic ladders in such spin-only constructions represent magnetic regions of a stripe pattern. The present model instead uses doped electronic ladders containing mobile charge. Moreover, its current analyses concern ground-state profiles and equal-time correlations, rather than a computed $S(\mathbf{q}, \omega)$. They can test the static antiphase relation in Eq. (3), but cannot yet demonstrate an hourglass spectrum or identify a neutron resonance.

2.4 Extended Hubbard ladders, materials, and transverse coupling

Material families and the limits of the ladder description

Magnetic ladder insulators establish the relevance of the ladder building block, but do not by themselves realize a doped electronic array. The original susceptibility and NMR comparison found a spin gap in the two-leg compound SrCu_2O_3 and a gapless response in the three-leg compound $\text{Sr}_2\text{Cu}_3\text{O}_5$ [23]. The approximately 420 K susceptibility gap quoted for SrCu_2O_3 is an energy expressed as Δ_s/k_B , not a superconducting temperature. Introducing mobile holes is a separate materials problem: chemical substitution can modify exchange paths, disorder, and carrier localization. Hole doping is possible in $\text{La}_{1-x}\text{Sr}_x\text{CuO}_{2.5}$, where an insulator-to-metal transition was observed without superconductivity down to 5 K in the reported measurements [40]. Its coupled-ladder electronic structure has substantial three-dimensional coupling [41]. Such compounds motivate studying coupled ladders and dimensionality, but do not identify our particular square or cubic kernels with a complete material Hamiltonian.

CaCu_2O_3 is a complementary warning against identifying a crystal motif with an isolated spin-gap ladder. Its buckled rung path has a Cu–O–Cu angle near 123° , and neutron diffraction finds incommensurate magnetic order below $T_N = 25$ K [42]. The distorted rung and competing couplings require a different magnetic description. Correlated electronic-structure calculations across SrCu_2O_3 , $\text{Sr}_2\text{Cu}_3\text{O}_5$, and CaCu_2O_3 likewise emphasize material-dependent hopping and exchange paths [43]. These are relevant structural counterexamples, not additional superconducting ladder realizations.

Within the chain–ladder family, $\text{La}_6\text{Ca}_8\text{Cu}_{24}\text{O}_{41}$ is a nominally hole-free magnetic reference, whereas $\text{La}_4\text{Sr}_{10}\text{Cu}_{24}\text{O}_{41}$ retains two formal holes per formula unit at O_{41} stoichiometry. The latter predominantly occupy chains, leaving approximately undoped ladders [24, 44]. This differs from both the intrinsically hole-bearing Sr parent and Ca-rich superconducting samples. Raman analyses also distinguish La_6Ca_8 from SrCu_2O_3 and $\text{Sr}_{14}\text{Cu}_{24}\text{O}_{41}$, finding different exchange anisotropies

within a common magnetic model [29]. The companion literature review provides individual annotations and a material-family comparison.

Pressure, carrier partition, and distinct transition scales

The superconducting chain–ladder cuprates provide a more direct motivation for the trellis extension. Uehara et al. reported superconducting onset temperatures of 12 K at 3 GPa and 9 K at 4.5 GPa in $\text{Sr}_{0.4}\text{Ca}_{13.6}\text{Cu}_{24}\text{O}_{41.84}$ [25]. The structure contains both Cu_2O_3 two-leg ladder sheets and CuO_2 chains. Neighboring ladders are offset longitudinally, producing the two zigzag paths of a trellis network [45]. Ca substitution is isovalent; it redistributes holes between chains and ladders rather than simply adding a known number of ladder holes. At O_{41} stoichiometry, the divalent Sr/Ca family has six formal holes per formula unit relative to Cu^{2+} , distributed over both subsystems. Polarized x-ray absorption directly distinguishes chain and ladder hole states [44]. Neither this total count divided by 24 nor the Ca fraction is the hole density per ladder Cu; oxygen nonstoichiometry further changes the charge count in samples such as Uehara’s $\text{O}_{41.84}$. NQR and pressure-dependent optical measurements likewise support pressure-induced transfer of holes into the ladders [46, 47]. Consequently, the model density $n = 15/16$ cannot be assigned directly to a Ca concentration or experimental pressure.

The established hydrostatic-pressure ladder transitions are on a roughly 10 K scale, well below the highest planar-cuprate transition temperatures; Hg-based planar cuprates reached a resistive T_c of 164 K under quasihydrostatic pressure [48]. This compares transition scales, not equivalent pressures or pairing energies. A separate preprint reports a 24 K resistive onset in $\text{Sr}_3\text{Ca}_{11}\text{Cu}_{24}\text{O}_{41}$ under about 0.06 GPa compression along c [49]. Its displayed magnetic transition near 6.8 K was obtained under a different loading direction and pressure, and therefore does not establish a bulk transition at 24 K. This qualified onset report is retained separately from the hydrostatic benchmark. An isolated ladder with short-range interactions has no finite-temperature superconducting transition: transverse phase coherence is essential, and local spin or pair gaps need not track the measured T_c . Pressure-dependent transverse transport supports the relevance of dimensional crossover, while changing filling and localization also matter [26].

The experimental charge-order landscape is richer than a monotonic conversion from a CDW to a superconductor. Transport, dielectric, and optical measurements found a CDW transition scale decreasing from about 210 K at $x = 0$ to 10 K at $x = 9$ [50]. Resonant x-ray scattering established a ladder hole crystal in $\text{Sr}_{14}\text{Cu}_{24}\text{O}_{41}$ and commensurate charge order also at $x = 10, 11, 12$ [27, 28]. High Ca content alone therefore does not establish the absence of CDW order. Composition, pressure, and commensurability must be specified when comparing a uniform paired model state with experiment.

Static magnetic order and dynamic spin correlations must also be separated. Neutron measurements on $\text{Sr}_{2.5}\text{Ca}_{11.5}\text{Cu}_{24}\text{O}_{41}$ found that the ordered moment decreased under pressure and appeared to disappear near the superconducting threshold [51]. A paired state without a resolved SDW is compatible with that trend, but vanishing static magnetization does not determine the spin gap. High-pressure NMR found gapped quasiparticle excitations in the superconducting state, described as s -wave-like in that specific sense [52]. This is not a phase-sensitive measurement of the relative signs of leg and rung pairing. A ladder’s d -wave-like bond structure need not force nodes on its Fermi surfaces; neither agreement nor disagreement of pairing symmetry follows from our bond signs alone. RIXS on Ca-substituted material further motivates including localization: the observed magnetic spectra were interpreted using off-ladder impurity potentials that immobilize holes [53].

Exchange anisotropy, orthorhombicity, and electronic parameters

Neutron scattering on $\text{Sr}_{14}\text{Cu}_{24}\text{O}_{41}$ separates a ladder spin gap near 32.5 meV from lower-energy chain-dimer excitations [54]. For $\text{La}_4\text{Sr}_{10}\text{Cu}_{24}\text{O}_{41}$, Notbohm et al. separate odd- and even-rung-parity scattering and fit a gap 26.4 ± 0.3 meV together with

$$J_{\text{leg}} = 186 \text{ meV}, \quad J_{\text{rung}} = 124 \text{ meV}, \quad J_{\text{cyc}} = 31 \text{ meV}. \quad (4)$$

The last coefficient multiplies their explicit four-spin combination; permutation-operator definitions can redistribute bilinear terms. Both neutron and optical analyses require attention to cyclic exchange [24, 29, 55]. The paper gives approximate electronic parameters $t_{\text{leg}} = 0.42$ eV, $t_{\text{rung}} = 0.34$ eV, and $U = 3.72$ eV, corresponding to $t_0/t \simeq 0.81$. The simpler leading-order relation gives

$$J_{\mu} \simeq \frac{4t_{\mu}^2}{U - V_{\mu}}, \quad \frac{t_0}{t} \simeq \sqrt{\frac{124}{186}} \simeq 0.82 \quad \text{if } V_{\text{rung}} = V_{\text{leg}}. \quad (5)$$

This conversion assumes a half-filled, strong-coupling, one-band description with the same virtual-charge denominator on both bonds. It is an approximate inference, not a direct neutron measurement of hopping. Higher-order processes and the multiorbital charge-transfer character limit its accuracy.

Orthorhombic symmetry permits inequivalent electronic directions but fixes neither the size nor the sign of this hopping anisotropy. Notbohm's stronger leg coupling occurs despite a longer leg Cu–Cu distance, demonstrating that the oxygen environment matters beyond bond length [24]. For planar cuprates, square connectivity is also distinct from isotropic couplings: in orthorhombic $\text{La}_{2-x}\text{Sr}_x\text{CuO}_4$ the crystallographic directions run along plaquette diagonals, and the calculated anisotropy appears in diagonal hopping [56]. A ladder already distinguishes rungs from legs without invoking a small planar distortion. No lattice-constant ratio or orthorhombic label selects a universal t_0/t .

The project Hamiltonian and material comparison

Each ladder has L rungs, two legs $\ell = 0, 1$, and $2L$ physical sites. Its Hamiltonian is

$$\begin{aligned} H_{\text{lad}} = & -t \sum_{x=1}^{L-1} \sum_{\ell, \sigma} (c_{x\ell\sigma}^{\dagger} c_{x+1, \ell\sigma} + \text{H.c.}) - t_0 \sum_{x, \sigma} (c_{x0\sigma}^{\dagger} c_{x1\sigma} + \text{H.c.}) \\ & + U \sum_{x, \ell} n_{x\ell\uparrow} n_{x\ell\downarrow} + V \sum_{\langle a, b \rangle_{\text{lad}}} n_a n_b. \end{aligned} \quad (6)$$

The final sum includes nearest-neighbor leg and rung bonds. Thus $V < 0$ is attractive in the density channel. Open boundaries are used along the ladder; $n = N/(2L)$ is the density per physical site and $p_h = 1 - n$ the hole density. The rung hopping t_0 and interladder hopping $t_{\perp}$ must not be confused with the next-nearest-neighbor hopping t' in two-dimensional Hubbard studies.

An approximate parameter scale comes from the optical analysis of $\text{Sr}_{14}\text{Cu}_{24}\text{O}_{41}$ by Popović et al. [45]: $t = t_0 = 0.26$ eV, interladder hopping $t_{xy} = 0.026$ eV, and $U = 2.1$ eV. Interpreting t_{xy} as the amplitude on each equivalent zigzag path gives $U/t \simeq 8.1$, $t_0/t = 1$, and $\tau_0/t = \tau_1/t = 0.1$, motivating our (8, 1, 0.1, 0.1) comparison. These are effective parameters from an optical model of the parent compound, not a pressure-resolved fit to Uehara's superconducting sample. That analysis does not independently determine the nearest-neighbor V used here; $V = 0$ is our baseline assumption.

There is independent motivation to vary V . The RIXS–DMRG comparison of Padma et al. favors an effective attraction $V/t \simeq -1$ to -1.25 for $\text{Sr}_{14}\text{Cu}_{24}\text{O}_{41}$ [30]. Their model uses $t = 0.38$ eV, $t_0/t = 0.84$, and an additional *intraladder* diagonal hopping $t'/t = -0.3$, distinct from our interladder τ_0, τ_1 . It adjusts U with V to hold the exchange scale fixed, whereas our scans hold $U/t = 8$. Thus its fitted attraction motivates extended interactions without providing a directly transferable parameter set for the present Hamiltonian. Pairing at $V = 0$ in our approximation would not establish that the material has no effective attraction. The complementary neutron–DMRG analysis of Ca-rich material uses $t = 0.4$ eV, the same 0.84 rung ratio and $-0.3t$ diagonal hopping in its author manuscript, and requires localization as well as attraction to explain the magnetic response [57]. The two studies concern different compositions and probes; neither directly measures the transverse phase stiffness or selects our mean-field ground state.

Table 1: Distinct material parameterizations in project hopping notation. The rows are not measurements of a universal parameter set. Material labels omit the common $\text{Cu}_{24}\text{O}_{41}$ suffix.

Source and material	t_0/t	Basis and indispensable qualifications
Notbohm: $\text{La}_4\text{Sr}_{10}$ [24]	$\simeq 0.81$	Approximate Hubbard extraction from magnetic spectra; cyclic exchange retained in spin fit; effectively undoped ladders.
Popović: Sr_{14} [45]	1	Optical model, $t = 0.26$ eV, $U = 2.1$ eV, two zigzag paths of 0.026 eV each.
Padma: Sr_{14} [30]	0.84	RIXS–DMRG model, $t = 0.38$ eV, diagonal $t' = -0.3t$, attraction and adjusted U .
Scheie: $\text{Sr}_{2.5}\text{Ca}_{11.5}$ [57]	0.84	Neutron–DMRG author-manuscript model, $t = 0.4$ eV, diagonal hopping, attraction, and pinning.

Table 1 motivates stronger-leg and isotropic comparisons. Our existing $t_0/t = 1.2$ – 1.4 scans instead explore stronger-rung physics. At 1.4, Eq. (5) would give $J_{\text{rung}}/J_{\text{leg}} \simeq 1.96$, opposite to the anisotropy in Eq. (4); those runs are controlled model studies, not fits justified by orthorhombicity. No numerical results at 0.81 or 0.84 are implied here. At finite U , solving the electronic Hubbard model already includes higher-order virtual processes, including cyclic exchange; adding a fitted spin-only ring term independently can double count contributions.

Chains, explicit oxygen orbitals, structural modulation, disorder, and longer-range Coulomb interactions are also absent from Eq. (6). In particular, correlated ab initio calculations find substantial longitudinal onsite-energy modulation in Ca-rich incommensurate structures [58]. The uniform model does not resolve that source of localization, even if its average hoppings are chosen from spectroscopy.

Transverse geometry and the mean-field reduction

The transverse reduction uses the positive scale

$$g = \frac{t_{\perp}^2}{|E_{\text{p}}|}, \quad E_{\text{p}}^{(h)} = E(N-2, S_z=0) + E(N, S_z=0) - 2E(N-1, S_z=\frac{1}{2}), \quad (7)$$

where negative $E_{\text{p}}^{(h)}$ denotes favorable hole binding. Geometry-dependent kernels convert anomalous, exchange, and density correlators into fields for the next ladder solve. The density is targeted through a chemical potential. The present ansatz retains real, spin-diagonal normal fields and

real anomalous fields; it does not provide unrestricted complex phases or noncollinear magnetism (Evidence 1).

For illustration, define $\gamma = 2g$. The same-leg pairing kernels, organized by the two leg indices, are

$$K_{\alpha}^{\text{square}} = \gamma \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad K_{\alpha}^{\text{cubic,u}} = \gamma \begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix}, \quad K_{\alpha}^{\text{cubic,f}} = \gamma \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \quad (8)$$

These matrices summarize one part of the full map, not the entire Hamiltonian. An equation-level comparison identifies the repulsive-ladder benchmark of Ref. [34] with the branch called *cubic frustrated* in this project. That name is the project's nomenclature. Its published result should not be assigned to all three geometries (Evidence 2).

The trellis adds a longitudinal displacement to the transverse map. Writing m for the ladder index, a minimal hopping Hamiltonian is

$$H_{\text{zigzag}} = - \sum_{m,i,\sigma} c_{m,i,1,\sigma}^{\dagger} (\tau_0 c_{m+1,i,0,\sigma} + \tau_1 c_{m+1,i+1,0,\sigma}) + \text{H.c.} \quad (9)$$

The symmetric case has $\tau_0 = \tau_1$; removing the shifted path recovers square connectivity in the one-ladder embedding. Triangular connections can frustrate interladder antiferromagnetic alignment, making this geometry a natural test of whether static stripe order remains favored over pairing. The outcome must be calculated with both channels present; it is not implied by frustration alone. Spin-model calculations predict momentum-dependent triplon signatures of this frustrated transverse exchange [59]. Electronic trellis Hubbard studies using variational clusters and cellular DMFT find superconducting domes, but disagree on a complex chiral solution in part of the studied regime [60]. This is a close theoretical comparison with different approximations and parameters. Our real-field ansatz cannot test complex order, and its zero-temperature amplitude is not a prediction of the experimental T_c .

In the same common-denominator approximation, let T_{mn} map sites on neighboring ladder n to ladder m , including both paths, and define $F_{ab} = \langle c_{b\uparrow} c_{a\downarrow} \rangle$ and $X_{ab}^{\sigma} = \langle c_{a\sigma}^{\dagger} c_{b\sigma} \rangle$. For the real fields retained here, the kernels take the form

$$\alpha_m = \frac{2}{|E_p|} \sum_{n \sim m} \mathcal{P} [T_{mn} (\mathcal{P} F_n) T_{mn}^T], \quad (10)$$

$$B_{m\sigma} = \frac{2}{|E_p|} \sum_{n \sim m} \mathcal{P} [T_{mn} \mathcal{P} (X_n^{\sigma} - \frac{1}{2} I) T_{mn}^T]. \quad (11)$$

Here $\mathcal{P}$ retains rung separations within the chosen range cutoff, and B combines diagonal density and off-diagonal normal fields. Each neighbor contributes separately; the two hops of a pair transfer are not split between different neighboring ladders. The shifted paths introduce mixed $\tau_0 \tau_1$ terms, so a coordination-number rescaling of Eq. (8) is insufficient. In particular, diagonal normal fields can depend on bond correlations as well as density (Evidence 15).

Reciprocity fixes the apparent directional asymmetry of the rung indexing. With $A = \tau_0 I + \tau_1 S$ and $S_{ij} = \delta_{j,i+1}$, the one-ladder map contains $AF^{00}A^T$ on leg 1 and $A^T F^{11}A$ on leg 0, with the range projections understood. Opposite shifts belong to opposite spatial neighbors at every iteration; alternating the indexing convention between iterations would instead alternate the map being solved. The comparison therefore uses a fixed reciprocal one-ladder cell and an explicit rectangular two-ladder A/B cell. Both are stationary spatial ansatzes, with different allowed transverse arrangements of longitudinal textures and different open end cuts. The two-ladder calculation targets the same mean density on each ladder separately; it does not allow unequal average fillings (Evidence 15).

These runs test whether geometry can change the selected order at fixed intraladder parameters and $V = 0$. A paired solution without resolved bulk CDW/SDW would be qualitatively compatible with a superconducting regime of the chain-ladder family, but would not locate it in pressure or reproduce the material’s whole phase diagram. The completed one- and two-ladder comparisons and their convergence limits are described in Section 3.12. The anomalous pairing amplitude is not a measurement of T_c , phase stiffness, or a dynamical spin gap.

This construction is conceptually related to coupled-ladder pairing mechanisms but does not implement every assumption of the original stripe proposals. In particular, the explicit alternating dopings in the period-four construction of Ref. [32] differ from an array of equivalent ladders, although that work also considers an equivalent-ladder period-two limit.

We study equivalent coupled ladders as a prescribed microscopic geometry. The observed stripe texture is an additional longitudinal charge and spin modulation; within the repeated-ladder embedding, its domain walls extend transverse to the ladder axes. This transverse alignment is a restriction of the embedding rather than an independently optimized arrangement of distinct ladders. Collective transverse quantum fluctuations are approximated by static mean fields.

3 Results and interpretation accumulated to date

3.1 Evidence hierarchy and common parameters

The results begin with the square-array anchor comparison at $t_0/t = 1.4$, $t_\perp/t = 0.1$, $U/t = 8$, and $n = 15/16$, with $V/t = 0$ and -0.4 , then extend to parameter grids, geometry and spatial-cell tests, and correlations within the stripe and paired groups. The coupled-state correlation data use $L = 64$ rungs and bond dimension $\chi = 200$. A separate $V = 0$ comparison tests $\chi = 400$, and the isolated-ladder backbone reaches $\chi = 1200$. These are different calculations with different error controls, rather than one uniformly converged data set.

It is useful to distinguish an observed texture, a numerically self-consistent state, stability against specified perturbations, and a thermodynamic ground state. Most trajectories provide texture and basin information without passing all self-consistency gates. The four square A/B runs in Section 3.14 do pass the fixed-point gates, but neither their acceptance nor the remaining finite-length and bond-dimension studies establish unrestricted stability or a thermodynamic phase. Here, *basin* describes access under a specified numerical map, not a measured free-energy basin.

3.2 The isolated ladder already has substantial pairing tendencies

The six-sector backbone at $V = 0$ and $t_0/t = 1.4$ gives the finite-system quantities in Table 2. All six final $\chi = 1200$ sectors pass their stored numerical gates. Changes from $\chi = 800$ are useful sensitivity observations, but not all sectors pass at that earlier stage; the comparison therefore does not constitute a complete bond-dimension extrapolation (Evidence 3, Evidence 5).

The measured pair correlations decay more slowly than the fitted charge correlations. Pair fits have R^2 between 0.953 and 0.982, while charge fits give 0.527–0.707. The quoted fit errors do not include the full finite-size, boundary, and truncation uncertainty. These results are consistent with a spin-gapped ladder with appreciable pairing correlations, but do not by themselves establish its asymptotic exponents or the ordered state of an array [21, 22].

Two other diagnostics illustrate the importance of conventions and fit quality. The stored charge parameter is 1.4247 with rung normalization and 0.7124 with site normalization. An unqualified

Table 2: Isolated-ladder evidence at $L = 64$, $U/t = 8$, $n = 15/16$, $t_0/t = 1.4$, and $V = 0$, from the final $\chi = 1200$ backbone. Negative binding energies use Eq. (7)’s convention.

Quantity	Value	Interpretation
Spin gap Δ_s/t	0.1533613	Fixed-particle-number spin excitation
Even-particle charge gap $\Delta_{c,2}/t$	0.01085830	Finite-size charge curvature
Hole binding $E_p^{(h)}/t$	-0.1465102	Favorable two-hole binding
Particle binding $E_p^{(e)}/t$	-0.1485427	Favorable two-particle binding
Pair decay exponent	0.7614 ± 0.0215	Fit to rung-pair correlations
Charge decay exponent	1.2158 ± 0.0830	Less reliable fit to charge correlations

symbol K_ρ would therefore be ambiguous. A central-charge fit yields approximately 2.009 with $R^2 = 0.155$ and is excluded from physical interpretation. The open-boundary density profile has a strong component at $q/\pi = 1/8$, which can arise as a Friedel response and is not independent proof of spontaneous charge order [61].

The Stage 1 equal-time covariance spectra are broad, with participation ranks around 48–63. The first bulk spin-odd candidate follows two edge-localized modes, and a charge mode at $q/\pi = 1/8$ first appears at rank 11. This motivated combining physically chosen fields with covariance-based directions instead of assuming that a handful of leading covariance modes would exhaust the relevant response. Equal-time covariance is itself distinct from a static susceptibility, which also depends on excitation energies.

3.3 Response discovery: a strong pairing direction with unresolved controls

The Stage 2 pilot reduces fourteen named fields to twelve independent probe directions: nine normal and three pairing directions. Its projected mean-field response has a leading square pairing multiplier of about 4.6688 and a leading positive normal multiplier of 1.1879. The cubic cases both give a leading pairing multiplier of 24.4730, while their leading normal behavior differs: the largest-magnitude normal multiplier is approximately -1.1879 for the frustrated construction and $+3.5637$ for the unfrustrated one (Evidence 4).

These values show that the isolated ladder supplies strong candidate pairing response and that transverse geometry changes the competing normal channels. They do not establish which nonlinear state wins. The discovery reciprocity error is approximately 0.121% and the normal/pair cross-block ratio 0.332%, but the maximum omitted-response norm fraction is 0.83267. Small errors within the projected matrix can therefore coexist with a large response outside the chosen subspace. The second-amplitude validation was not present in the reviewed evidence (Evidence 5).

More fundamentally, the zero-field isolated ladder is not a fixed point of the full normal mean-field map: its square-map image has a normalized field norm of approximately 0.0306655, dominated by normal fields. A derivative at that reference is a bare-ladder diagnostic, not the stability matrix of a self-consistent dressed normal state. The associated square estimate $t_\perp/t \simeq 0.04628$ must consequently remain a projected bare-reference onset estimate, not a phase boundary.

This resolves the apparent tension between a large bare pairing response and the later nonlinear stripe trajectories. They concern different reference states, finite-amplitude feedback, and restricted response information. Positive or negative multipliers describe growth or alternation of the specified

iteration. Neither is a physical excitation frequency or a global energy criterion. A physical two-sublattice interpretation of a raw two-cycle also requires a mapping to the transverse lattice and the appropriate partner-field energy [35].

3.4 Why the earlier paired-state interpretation was revised

The loose six-seed $V = 0$ campaign initially produced six software-accepted, pairing-bearing endpoints, with a corrected energy spread of only $6.0274 \times 10^{-7}t$ per physical site. This suggested a common paired region, but the later history analysis showed that it did not establish a unique stable paired basin. Three histories stopped after six evaluations; three others showed coherent spin-density growth during raw iteration and subsequent suppression under Anderson acceleration. The conspicuous change in one trajectory is reproduced by its first Anderson coefficients, $+22.6811$ and -21.6811 (Evidence 7).

This finding does not imply that Anderson acceleration changes the fixed-point equation or that every accelerated solution is unphysical. It shows that short histories and accelerated access do not test stability against a resolved competing-order perturbation. The subsequent 95%/5% two-family experiment addresses that question by using representative paired and striped reference fields, adding a finite component of the competing order, and retaining raw updates throughout.

A related warning comes from the square $(t_0/t, V/t) = (1.0, -0.4)$ coverage point. Its *diverging* outcome is a convergence classification: the fields remain bounded while pairing gives way to a stripe-like texture and an accelerated residual spike triggers the stop. It is not evidence of a divergent physical observable or an accepted phase (Evidence 8).

3.5 Attraction: two contrasting starts approach a common paired state

At $(t_0/t, V/t) = (1.4, -0.4)$, both original 80-evaluation trajectories approach the same pairing-dominated state. The initially strong stripe's bulk leg-odd spin Hartree RMS decreases from about $1.99 \times 10^{-2}t$ to $3.88 \times 10^{-8}t$; the pairing start ends near $2.49 \times 10^{-8}t$. Both pairing-field RMS values settle near $2.8236 \times 10^{-3}t$. Bulk RMS here uses rungs 6–59 (Evidence 9).

The physical symmetrized anomalous correlators have bulk leg amplitudes near $+0.035217$ and rung amplitudes near -0.053242 , establishing a d -wave-like internal sign pattern in the stored convention. These are not normalized versions of Eq. (1) or pair-pair correlators. The full anomalous matrices agree to approximately 5.85×10^{-5} in relative L2 norm, and the largest site-density difference is below 6.23×10^{-6} . The residual boundary density profile remains visible; its survival does not indicate a surviving strong static spin stripe.

The terminal corrected energies differ by only $7.49 \times 10^{-9}t$ per site, supporting agreement of the two trajectories at their resolved accuracy. Both historical acceptance flags remain false. The principal issues are residual fluctuations in nearly vanishing channels, an oversensitive scalar slow-mode check, and a marginal energy-window miss in one lineage. The later noise-aware history checks support the plateau interpretation, but do not retroactively certify every original gate. Thus the result is strong evidence for a common paired attractor at these numerical settings, without proof of a global minimum or higher- χ stability.

3.6 Zero attraction: stripe order grows while anomalous pairing recedes

The corresponding $V = 0$ trajectories behave differently. The stripe start reaches 80 evaluations with bulk leg-pairing field RMS $1.75 \times 10^{-10}t$ and spin Hartree RMS $0.02414t$. The pairing start

ends at the wall-time limit after 62 evaluations: its pairing RMS has fallen from $0.004502t$ to $0.0007090t$, while spin RMS has grown from $0.001816t$ to $0.02251t$. Pairing falls by another 65.6% across the final five records. The trend is already present before the deadline-limited final density solve (Evidence 10).

The residual anomalous correlators in the pairing-start trajectory still have opposite leg/rung signs, but their amplitudes are decreasing rapidly. They do not establish settled coexistence. The two spin textures are similar but not identical: the bulk physical spin profiles differ by 26.4% in relative L2 norm, and the density difference can reach 0.04365 per site. The terminal global relative residuals are about 5.88×10^{-4} and 2.65×10^{-2} for the stripe and pairing starts, respectively, against a 10^{-4} threshold.

The working interpretation is that a robust stripe texture has been identified and that the finite competing-order perturbation drives the pairing-dominated start toward it under raw iteration. Strict spatial self-consistency remains pending. In particular, the stripe amplitude's apparent saturation does not eliminate slow rearrangement of the profile. These observations support strong competition between static spin/charge order and the anomalous pairing solution at this point. They do not prove that the final stripe is the global minimum or that all local pairing correlations have disappeared.

3.7 Stripe wavevectors and slow texture relaxation

Both $V = 0$ endpoint profiles have dominant full-length Fourier modes $m_c = 4$ for charge and $m_s = 30$ for leg-odd spin. With $q_m = 2\pi m/L$ and $L = 64$,

$$q_c = \frac{\pi}{8}, \quad q_s = \frac{15\pi}{16}, \quad \kappa = \pi - q_s = \frac{\pi}{16}. \quad (12)$$

The observed relation $q_c = 2\kappa$ and the sign reversals of the staggered spin envelope near charge-depleted regions support an antiphase stripe interpretation. The charge period is 16 rungs and the spin-envelope period 32 rungs.

At hole density $p_h = 1/16$, the magnetic displacement in reciprocal-lattice units is $\delta_s = \kappa/(2\pi) = 1/32 = p_h/2$. It differs from the often discussed cuprate relation $\delta_s \simeq p_h$. In an ideal domain-wall counting picture with stripe filling ν holes per unit transverse length, $p_h = \nu/d$ and $\delta_s = 1/(2d)$; the present wavevectors would correspond to $\nu = 1$, rather than the $\nu = 1/2$ used in the usual half-filled-stripe cartoon. This is a conditional interpretation of the finite-system wavelength, not a measurement of wall filling. Neither the preferred wavelength nor its scaling with doping has been established.

The September 13 follow-up explains why the stripe looks settled while failing strict convergence. Its bulk spin RMS changes by only 0.0528% over the last twenty records, but more than 99.999% of the squared final spin residual is orthogonal to an overall rescaling of the applied spin field. Approximately 99% of that residual power lies in the bulk. The dominant remaining error concerns spatial shape.

For example, interpolated crossings of the staggered spin envelope near the third and fourth walls shift from 39.3999 and 55.1830 at evaluation 60 to 39.5165 and 55.2673 at evaluation 80. Unequal shifts reveal texture rearrangement. They are sub-rung profile diagnostics, not measured motion of physical objects. The stopping diagnostic estimates a contraction near 0.99059, suggesting very slow relaxation if that behavior persists; the extrapolation is neither a certified error bound nor a physical soft-mode eigenvalue (Evidence 10).

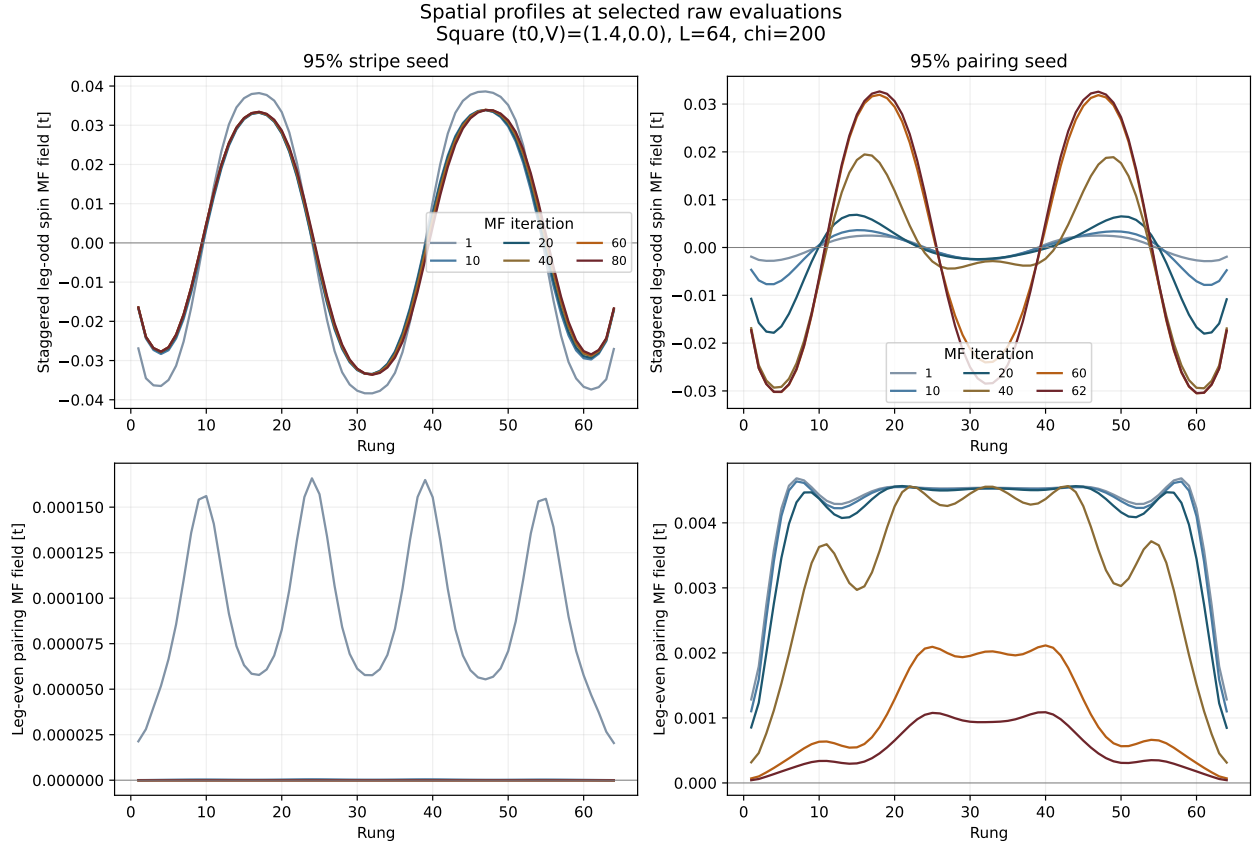


Figure 1: Previously generated spatial-history figure for the two $V = 0$ anchor trajectories. The upper row shows the staggered leg-odd spin mean field; the lower row shows the leg-even pairing mean field. The two lower panels use different vertical scales. The stripe start loses pairing, while the pairing start develops a stripe texture and its pairing decreases. Curves are self-consistency evaluations, not physical time evolution. The final states remain unaccepted for resolved field drift. Source: [Evidence 10](#).

3.8 Energy comparisons and what the bond-dimension check adds

The relevant energy is the canonical variational functional, including the mean-field double-counting contribution. If L_{lin} denotes the expectation of the applied linear mean-field terms,

$$E_{\text{var}} = E_{\text{eff}} + \mu N + (E_{\perp} - L_{\text{lin}}) = \langle H_{\text{lad}} \rangle + E_{\perp}. \quad (13)$$

The density-targeted diagnostic is $E_{\star} = E_{\text{var}} + \mu(N_{\star} - N)$, a leading correction to the target density. It does not remove curvature and optimization errors. Effective-Hamiltonian eigenvalues alone cannot rank the branches (Evidence 1).

At $V = 0$, the separate matched $\chi = 400$ comparison gives a stripe endpoint lower than the paired endpoint by $2.4336 \times 10^{-4}t$ per physical site, or $0.03115t$ on the 128-site ladder. The September 6 working decision treated the available evidence as sufficient to regard the advantage as reasonably robust over the tested $\chi = 200$ to 400 increase. Neither original endpoint satisfies all formal acceptance gates, so the value remains a terminal diagnostic difference rather than a fully certified fixed-point ranking (Evidence 6).

The newer raw $\chi = 200$ trajectories instead end with a separation $8.99 \times 10^{-5}t$ per site while the pairing lineage is still evolving. This is not an independent estimate of the same equilibrium energy gap. Similarly, an early low energy on an unconverged trajectory cannot be selected as a lower-energy phase. During startup, the functional combines the applied partner fields with current correlations; raw self-consistency iteration is not guaranteed to minimize a single self-consistent state's energy monotonically.

3.9 Complete square and unfrustrated-cubic grids

The September 18 review extends the anchor comparison to $t_0/t = 1.0, 1.2, 1.4$ and $V/t = 0, -0.2, -0.4$, with $U/t = 8$, $n = 0.9375$, $t_{\perp}/t = 0.1$, $L = 64$ rungs and $\chi = 200$. The two starts are reciprocal 95%/5% mixtures of the reference stripe and paired correlations, reconstructed with the target couplings. They evolve with raw MF updates, without Anderson acceleration. All phase descriptions below refer to finite-run physical correlations, not certified minima (Evidence 11).

Figure 2 places the two phase diagrams side by side with the same coordinate axes. After including both $(1.4, 0)$ continuations, the coarse square grid has seven stripe coordinates and two paired coordinates, $(1.4, -0.4)$ and $(1.4, -0.2)$. Both seed families now reach the same qualitative basin at each coordinate. At $(1.4, 0)$ the stripe and pairing lineages contain 100 and 82 cumulative evaluations, respectively; the late pairing collapse seen in the earlier history is therefore followed by a stripe endpoint. The whole square grid contains 902 evaluations across 18 lineages, including 20 source jobs, with no formally accepted endpoint (Evidence 12).

In contrast, all 18 unfrustrated-cubic starts reach essentially unpaired CDW/SDW stripes after 60 evaluations. Physical leg-odd spin RMS ranges from 0.280 to 0.356, while the largest leg-pair RMS is 2.21×10^{-11} . The dominant charge and spin modes are $m = 4$ and 30, corresponding to nominal 16-rung charge and 32-rung staggered-spin-envelope periods. The two square paired coordinates are striped in cubic. Every cubic endpoint passes its final energy-window, density, inner-DMRG and identity checks; remaining field/profile and slow-mode checks prevent acceptance. This is evidence of geometry-dependent nonlinear basin selection, not an energetic ranking of different Hamiltonians or an exclusion of every other cubic basin.

The full energy histories in Fig. 3 retain the startup and delayed conversions. In particular, square $(1.2, -0.4)$ has an extended pairing-seed energy plateau before its late decrease; square $(1.4, 0)$ evolves slowly into its continuation. Cubic energy histories generally settle earlier. All

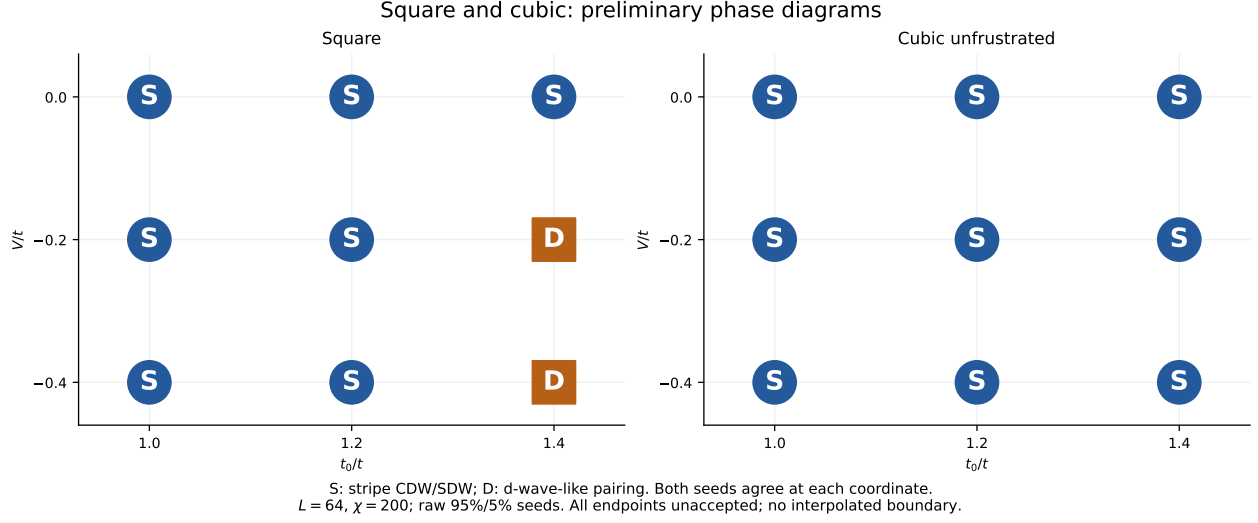


Figure 2: Preliminary square (left) and unfrustrated-cubic (right) phase diagrams. Both seeds reach the same observed family at every coordinate: seven stripe and two paired points on square, nine stripe points in cubic. S denotes CDW/SDW stripes and D opposite-sign leg/rung pairing. All endpoints remain unaccepted at $\chi = 200$; no boundary is interpolated. Sources: [Evidence 12](#) and [Evidence 11](#).

plotted energies use the stored canonical functional with target-density correction, and each panel keeps its own vertical scale. The comparison is between numerical trajectories under different geometries, not a ranking of their absolute energies as phases of one Hamiltonian. Early off-self-consistent energy dips and late flat energies do not establish stationary minima.

Figures 4 and 5 show the complementary physical spin and pairing RMS histories. They use the same definitions and logarithmic scales in both geometries, unlike the older standalone square plots of MF-field amplitudes. Spin is the RMS of the physical leg-odd spin on rungs 6–59; pairing is the physical leg-even nearest-neighbor leg-pair RMS on bonds 6–58. Square spin decays while pairing survives at $(1.4, -0.4)$ and $(1.4, -0.2)$; cubic instead retains strong spin and loses pairing at both coordinates. At square $(1.2, -0.4)$ and $(1.4, 0)$, delayed pairing collapse accompanies increasing spin. The long transient in these panels explains why a short, flat energy history can be misleading. Tiny pairing fluctuations after collapse are numerical remnants, not resolved renewed order.

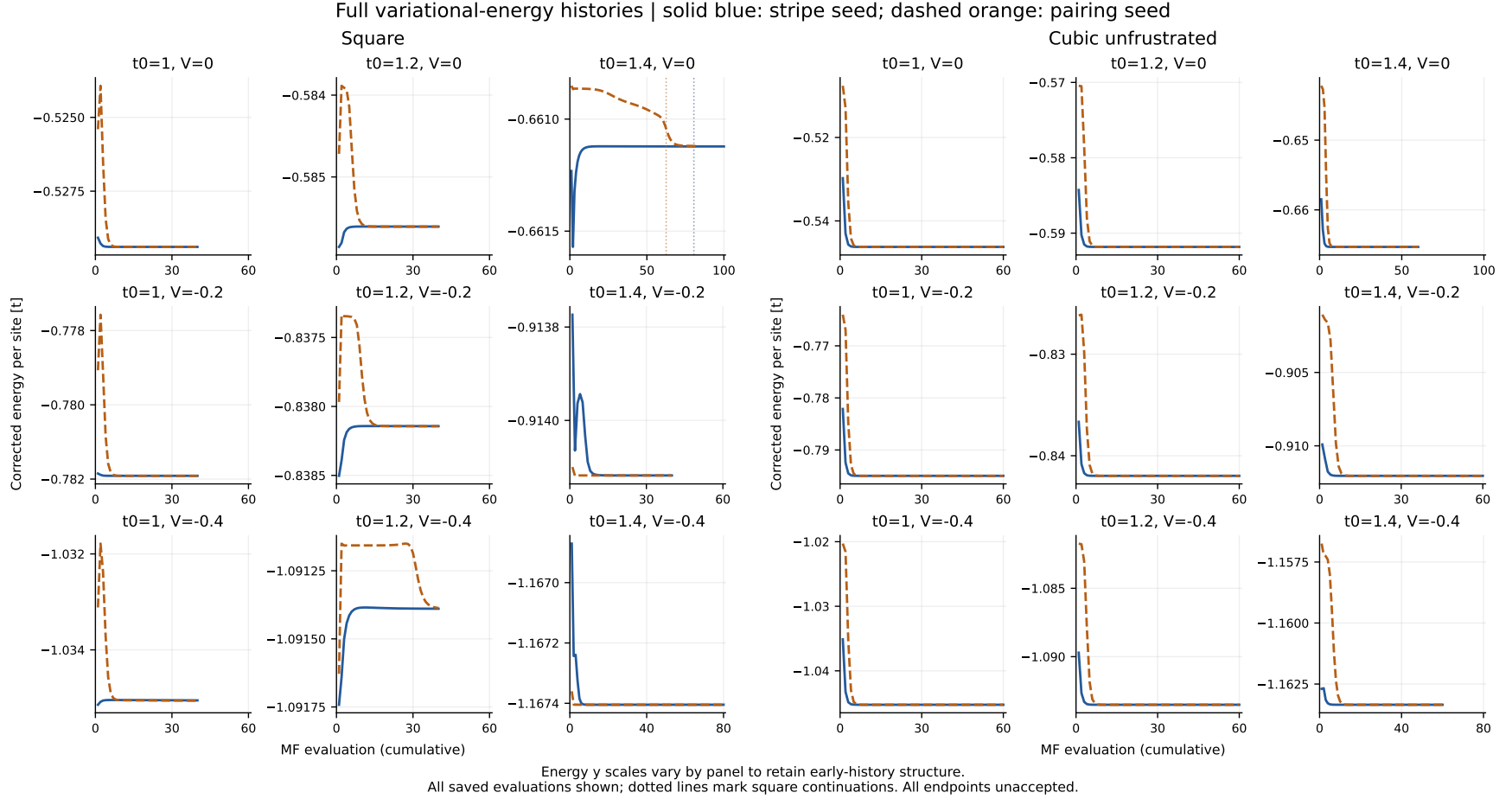


Figure 3: Full target-corrected canonical energy grids, square left and cubic unfrustrated right. Within each geometry, columns are $t_0 = 1, 1.2, 1.4$ and rows are $V = 0, -0.2, -0.4$. Solid blue and dashed orange denote stripe and pairing seeds. All 902 square and 1080 cubic saved evaluations are included. Dotted lines mark the square continuations after evaluations 80 and 62; curves end at their own final evaluations. Matching coordinates share an iteration range, while energy y scales vary by panel.

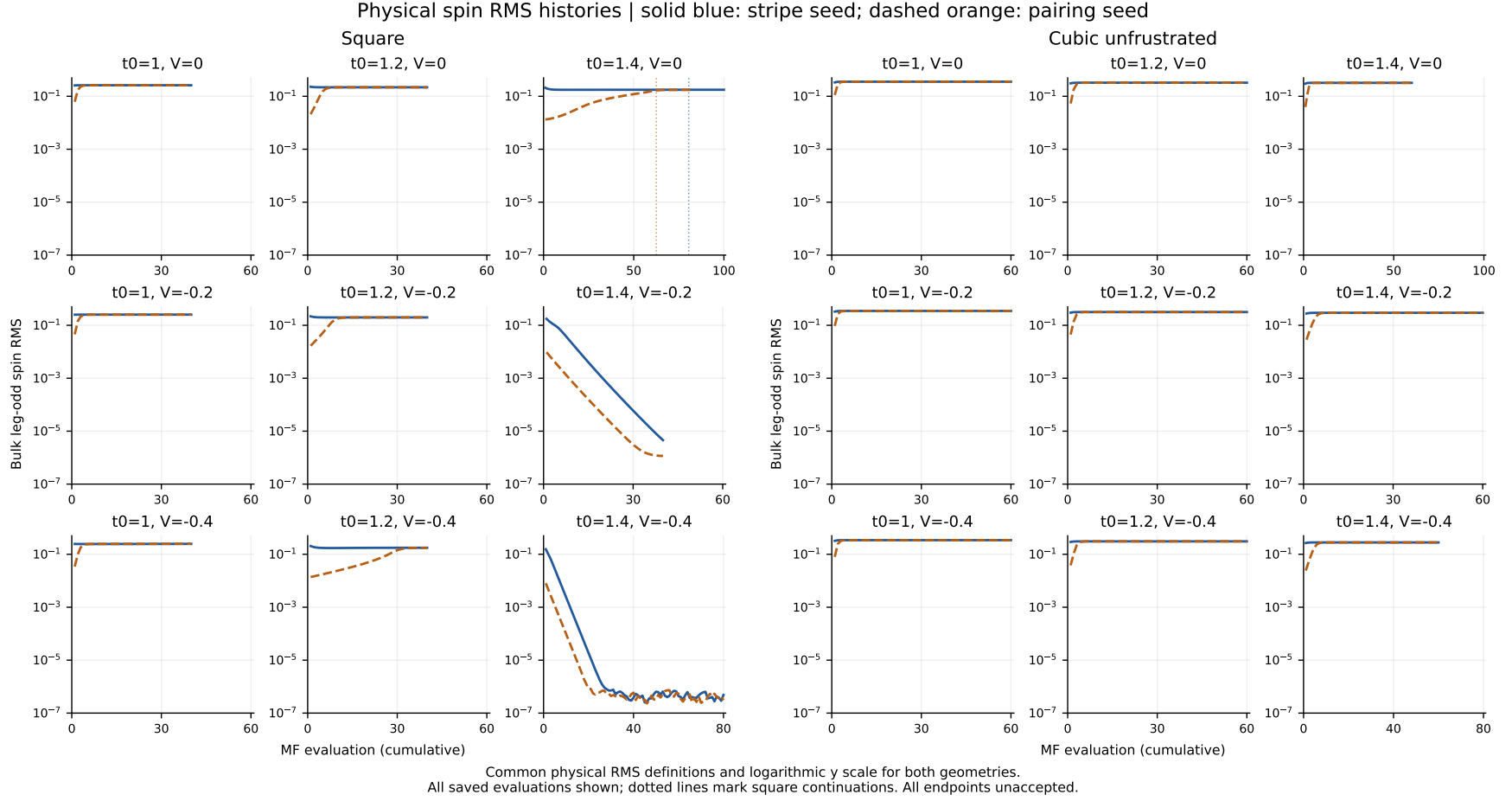


Figure 4: Physical leg-odd spin RMS histories, square left and cubic right, over rungs 6–59. All panels use a common logarithmic y scale. The point order, seed colors and continuation markers match Fig. 3. These are physical correlations, not geometry-dependent Hartree-field amplitudes. Spin decays at the two square paired coordinates but remains strong throughout cubic.

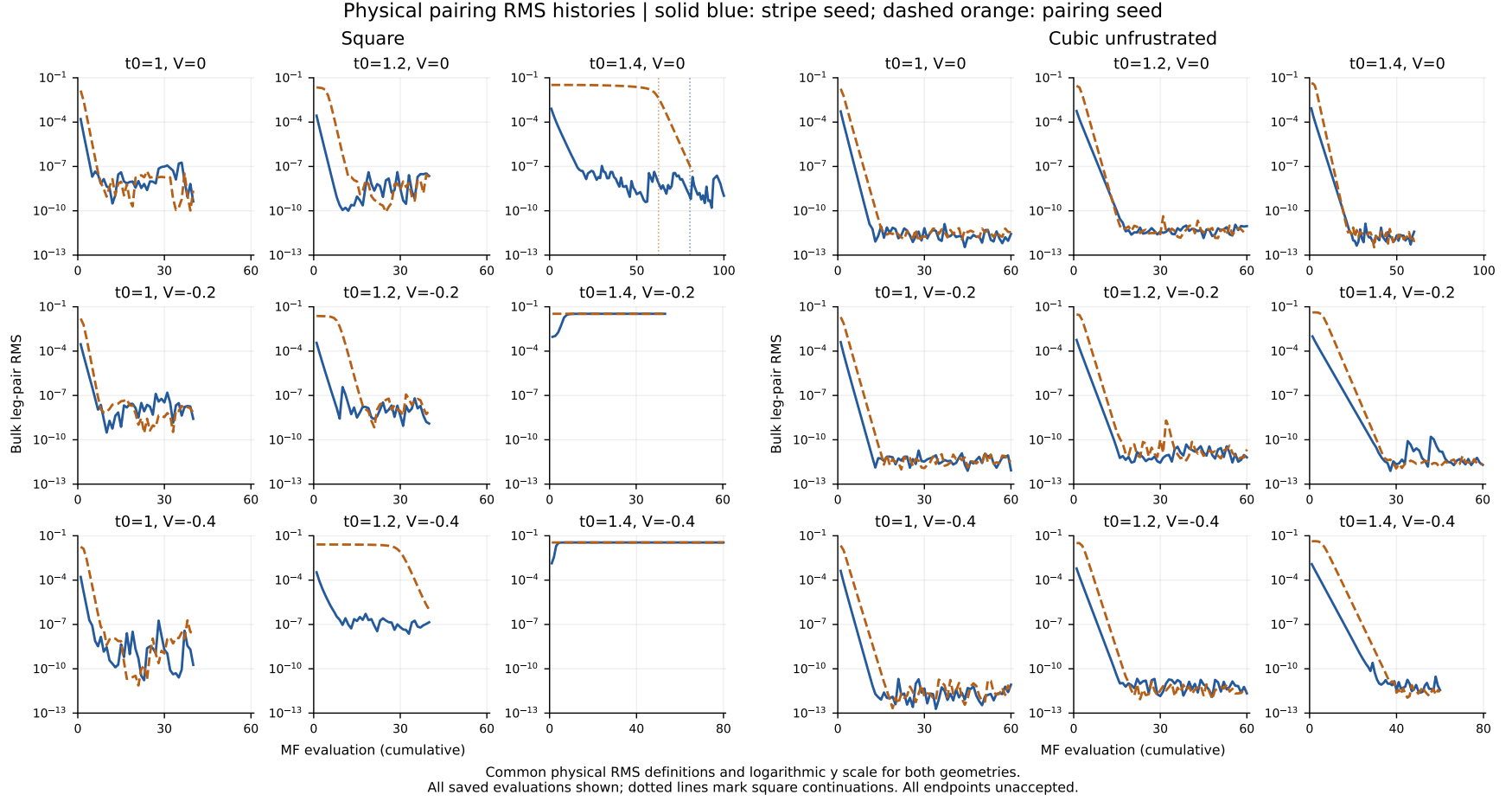


Figure 5: Physical leg-pair RMS histories on bonds 6–58, square left and cubic right, with common logarithmic y scale. The layout matches Fig. 4. Pairing survives at the two square paired points and decays to negligible amplitudes at every cubic point. The full histories retain delayed square pairing collapse, including the $V = 0$ continuation. None of the endpoints is formally accepted.

3.10 Finer square cuts and the shape of the variational energy

The finer cuts add $V/t = -0.15, -0.10, -0.05$ at $t_0/t = 1.4$ and $t_0/t = 1.25, 1.30, 1.35$ at $V/t = -0.4$, with 60 raw evaluations per seed. Both seeds reach paired trajectories at four new points. Distinct stripe and paired trajectories survive at $(1.25, -0.4)$ and $(1.4, -0.05)$. Their signed endpoint energy differences, pairing seed minus stripe seed, are -3.189×10^{-5} and $-5.952 \times 10^{-5} t/\text{site}$. These exceed the sums of the corresponding final-ten energy ranges, 1.574×10^{-7} and $7.461 \times 10^{-8} t/\text{site}$, but recent drift does not bound the remaining convergence error. All twelve endpoints remain unaccepted (Evidence 13).

Figure 6 shows the full stored variational energy along both cuts, including the coarse endpoints and the latest $V = 0$ continuations. The quantity is the canonical functional with its field-dependent double-counting terms and target-density correction,

$$e_{\text{target}} = E_{\text{var}}/N_s + \mu(n_{\text{target}} - n), \quad N_s = 2L = 128. \quad (14)$$

It is not the effective-Hamiltonian eigenvalue. No additional field-independent perturbative offset has been introduced. On this scale the two seed curves nearly overlap: the V cut is almost linear, while the t_0 cut has gradual curvature. Curves identify the initial seed, not a persistent phase, because independent runs can change their outcome as the parameter changes.

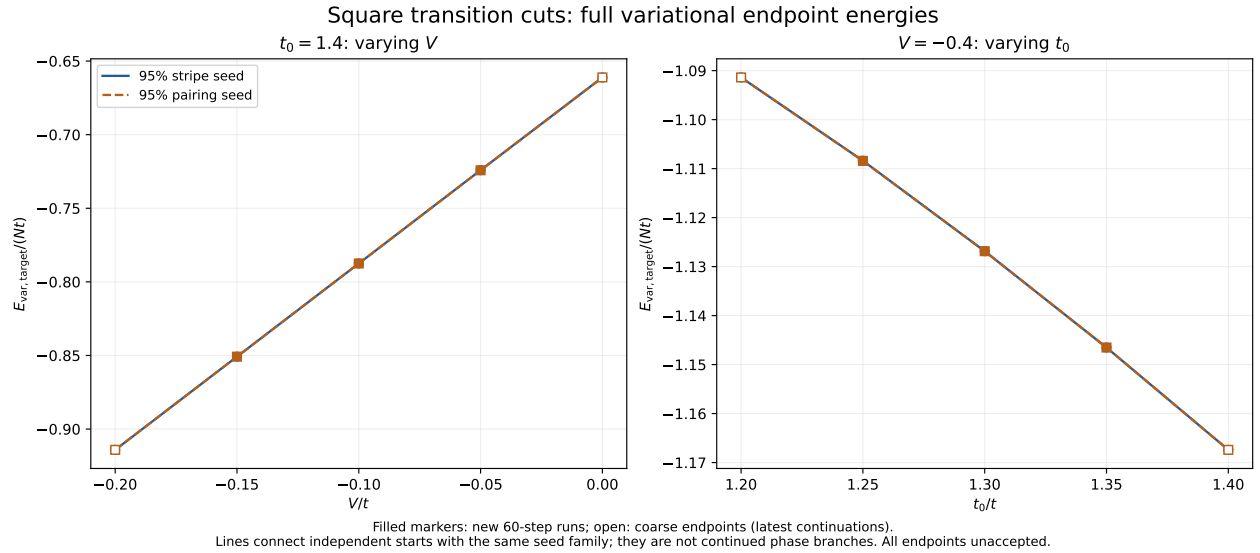


Figure 6: Full canonical variational endpoint energies, corrected to the target density. Filled points are the new 60-step starts; open points use the coarse endpoints and latest continuations. Lines guide the eye between independent starts; they are not parameter-continuation branches. Total energy equals the displayed energy per site times 128.

To resolve small shape changes, Fig. 7 subtracts one common linear chord from both seed curves on each cut. Its endpoints are the mean seed energies at the two outer coordinates. This preserves both seed gaps and changes in slope. The lower panels display adjacent secants $\Delta e/\Delta p$, with $p = V$ or t_0 , rather than a fitted derivative. Along the pairing-seed V curve the slopes are approximately 1.266231, 1.266144, 1.266212 and 1.261367. Thus the last interval, $-0.05 \leq V/t \leq 0$, has a slope smaller by about 0.38% than its predecessor. The stripe-seed curve also bends downward in that interval. This small feature lies near the change from paired to stripe outcomes and is compatible with a weak first-order crossing, but does not resolve one.

For a first-order transition the minimizing energy can remain continuous while its first derivative jumps. Five sampled coordinates do not distinguish that kink from a continuous change of curvature. On the t_0 cut the pairing-seed secants, -0.340515 , -0.368857 , -0.393241 , -0.417749 , change nearly regularly, with no obvious sharp cusp at this resolution. The broad arch after linear subtraction is not itself a singularity. Absence of a visible kink does not exclude a weak first-order transition.

The two distinct textures at one point on each cut make metastability and first order worth testing. However, these are not two stationary phase branches traced through a crossing, nor parameter-sweep hysteresis. The signed E_p values at the new points were linearly interpolated between the coarse endpoints as requested; $g = t_\perp^2/|E_p|$ therefore also changes along each cut. The plotted slopes include this coupling dependence and cannot be identified with the bare V or rung-hopping conjugate observable alone. Coarse and fine runs also have different stopping controls.

The weak spin at the paired $(1.4, -0.05)$ endpoint requires a spatial qualification. Its usual spin RMS on rungs 6–59 rises 1.50% over the last ten evaluations to 0.005606, but 96.7% of its spin-squared weight lies in the outer 14 rungs at each end. Central 32-rung spin still decreases, while leg-pair RMS remains 0.032991. This is not established bulk coexistence or a demonstrated bulk return to stripes. At paired $(1.25, -0.4)$, the spin remainder decreases by 52.1% over the final ten evaluations.

Figure 8 places spin and pairing alongside the energy cuts. Coarse and fine points use the same physical definitions: $M_i = (\langle S_{i,0}^z \rangle - \langle S_{i,1}^z \rangle)/2$, with spin RMS over rungs 6–59, and symmetric nearest-leg anomalous pairing averaged over the two legs before its RMS on bonds with left rungs 6–58. These are not source-field norms or four-fermion pair-pair correlations. Central-window spin exposes the end-weighted remainder at $(1.4, -0.05)$. Pairing stays finite almost up to the sampled change of outcome, while the stripe-seeded amplitudes change at a different sample. This motivates testing competing finite-amplitude branches, but joining independent starts does not measure a discontinuity, coexistence interval or hysteresis. A logarithmic companion in the report resolves the small remainders.

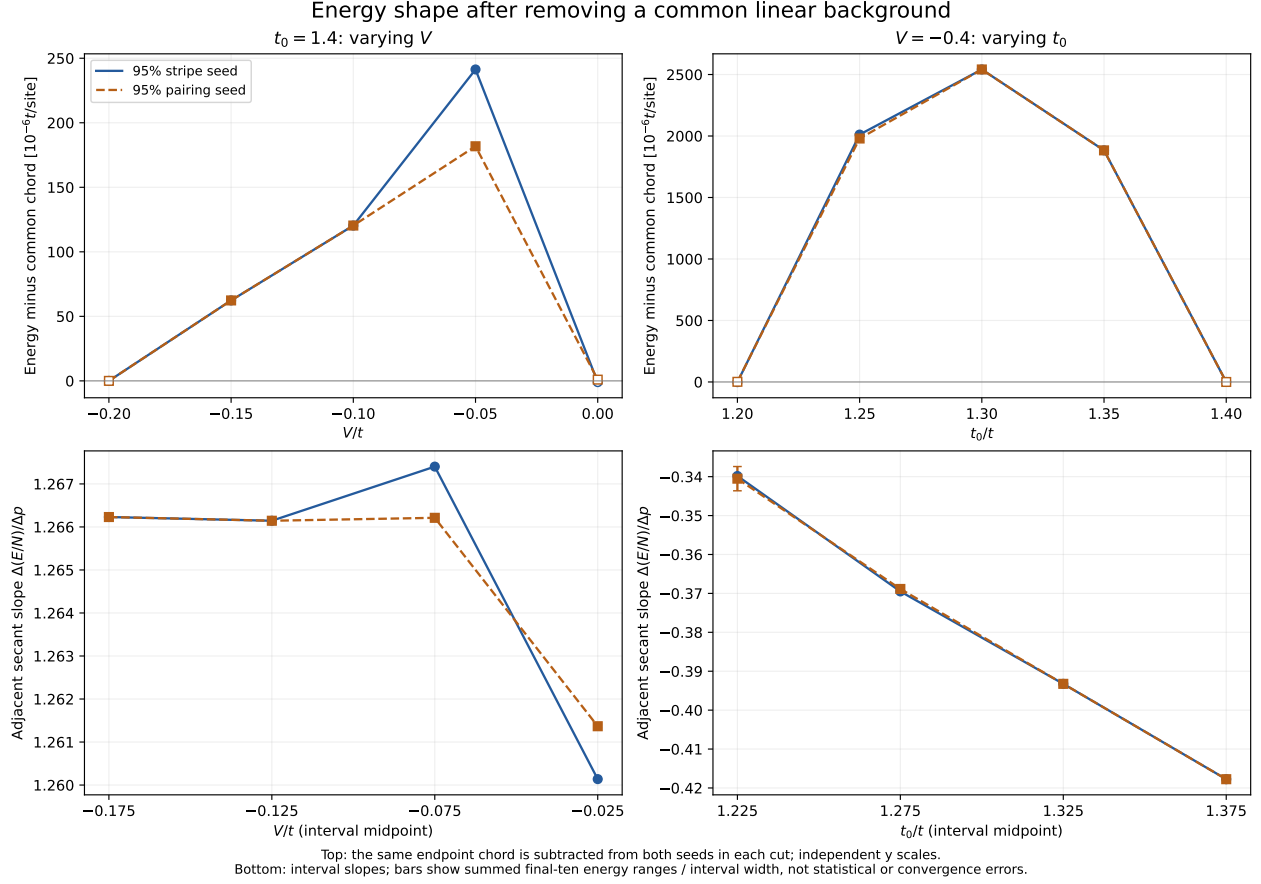


Figure 7: Energy shape and adjacent-interval slopes. The same chord is subtracted from both seed curves within each cut; the two cuts have independent vertical scales. Bars on the secants are the sum of endpoint final-ten energy ranges divided by interval width, not statistical or convergence error bounds. In particular, the coarse $t_0 = 1.2$ pairing lineage still has an energy range $1.56 \times 10^{-4} t/\text{site}$. Polygonal corners between samples are not measured derivative jumps.

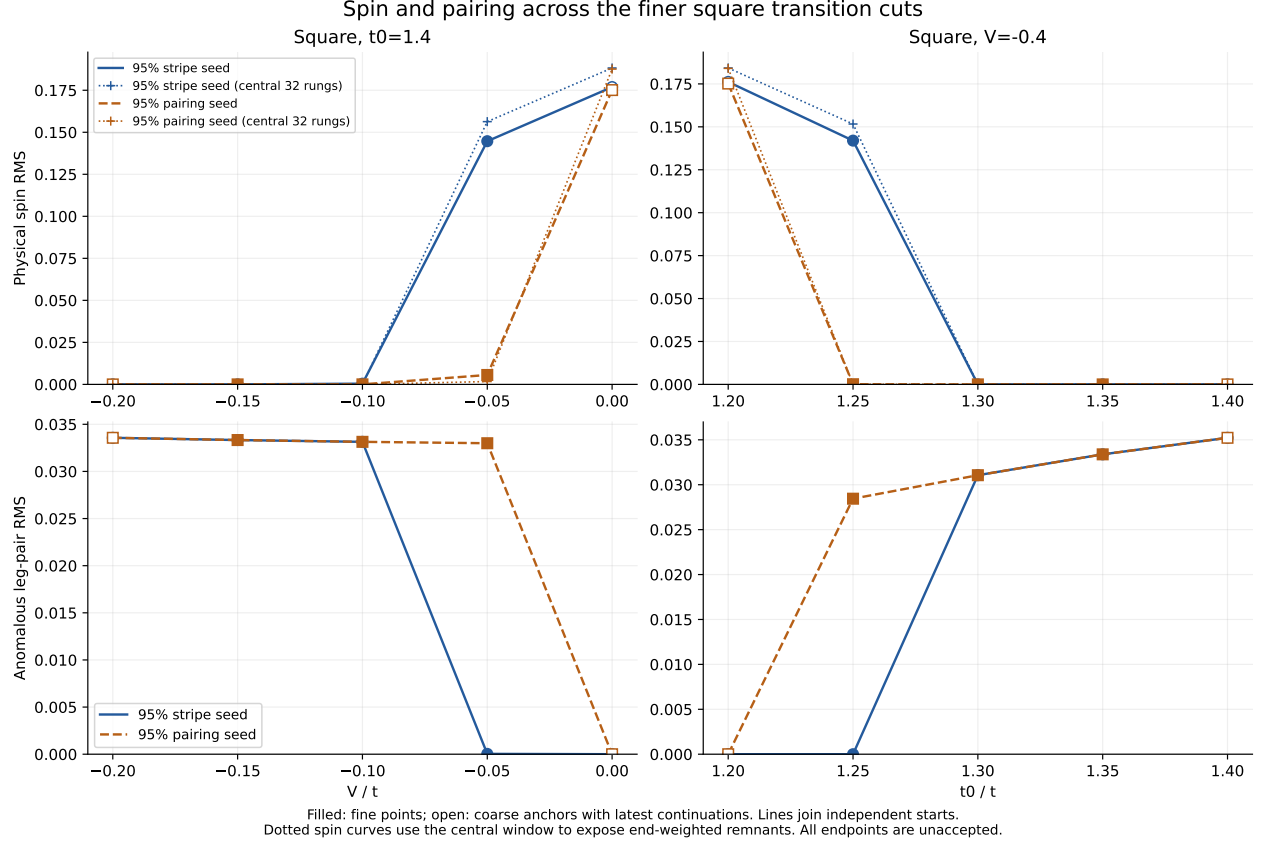


Figure 8: Spin and anomalous leg-pair RMS along both finer square cuts, using the same twenty endpoints as Fig. 6. Filled markers denote fine starts; open markers denote coarse anchors with latest continuations. Dotted spin curves use central rungs 17–48. Lines join independent finite-iteration starts, not stationary phase branches. All endpoints remain unaccepted.

3.11 Repulsive interaction on square: loss of pairing and magnetic defects

All four square $(t_0/t, V/t) = (1.2, +0.2)$ starts now complete 60 raw evaluations at $L = 64$, $\chi = 200$: the two 95%/5% reference mixtures and intertwined patterns with initial charge wavelengths eight and sixteen. The additional seeds place pairing near hole-rich regions and spin nodes, testing whether the legacy frustrated-cubic texture survives on square. Figure 9 shows that all four lose anomalous pairing. Stripe, pairing and period-16 starts end at spin RMS 0.23321–0.23358 and leg-pair RMS below 8.3×10^{-10} . The period-eight start ends at spin RMS 0.20700 and leg-pair RMS 7.01×10^{-9} , with the latter falling another 87.8% over the last ten evaluations. None retains appreciable anomalous coexistence (Evidence 14).

The shorter-wavelength start has not merged with the other three. Figure 10 shows four spin-envelope nodes in the stripe, pairing and period-16 endpoints, near rungs 10, 25, 40 and 55. Their charge/spin Fourier bins are $m = 4/30$, giving nominal charge and spin-envelope periods 16/32. The period-eight start evolves from eight nodes at evaluation one to six near 10.16, 23.21, 26.81, 38.30, 41.89, 54.91. Two close node pairs survive around the inner hole-rich regions. Charge still has dominant $m = 4$, while the spin spectrum has comparable $m = 31, 30, 28$ amplitudes 0.09264, 0.08450, 0.06761. Its largest bin should not be interpreted as a clean new spin wavelength. The spatial evidence is consistent with incomplete coarsening or a trapped magnetic defect pattern, rather than a surviving period-eight intertwined state.

The period-eight energy is $-0.332811625875 t/\text{site}$, versus approximately $-0.333744 t/\text{site}$ for the other three. Its excess over the stripe-seeded endpoint is $9.3242 \times 10^{-4} t/\text{site}$; their summed final-ten energy ranges are $8.96 \times 10^{-8} t/\text{site}$. This is a useful diagnostic distinction, not a certified metastable branch energy. All four fail global field/slow-mode and spatial charge, spin and exchange checks. Maximum pointwise spin changes over the last ten evaluations remain 0.00358–0.00655, despite nearly constant RMS. Stripe and period-eight starts also fail an inner-DMRG window; period sixteen fails the energy window. Flat energy and a steady norm conceal moving walls.

The completed wavelength test strengthens the qualification on the earlier physical interpretation: repulsive V alone does not guarantee intertwining across these geometries. Other square parameters or basins remain possible. Loss of anomalous order also does not determine whether connected pair–pair correlations survive in the stripes. The completed retrospective measurements address that separate question in Section 3.13, without changing these acceptance flags.

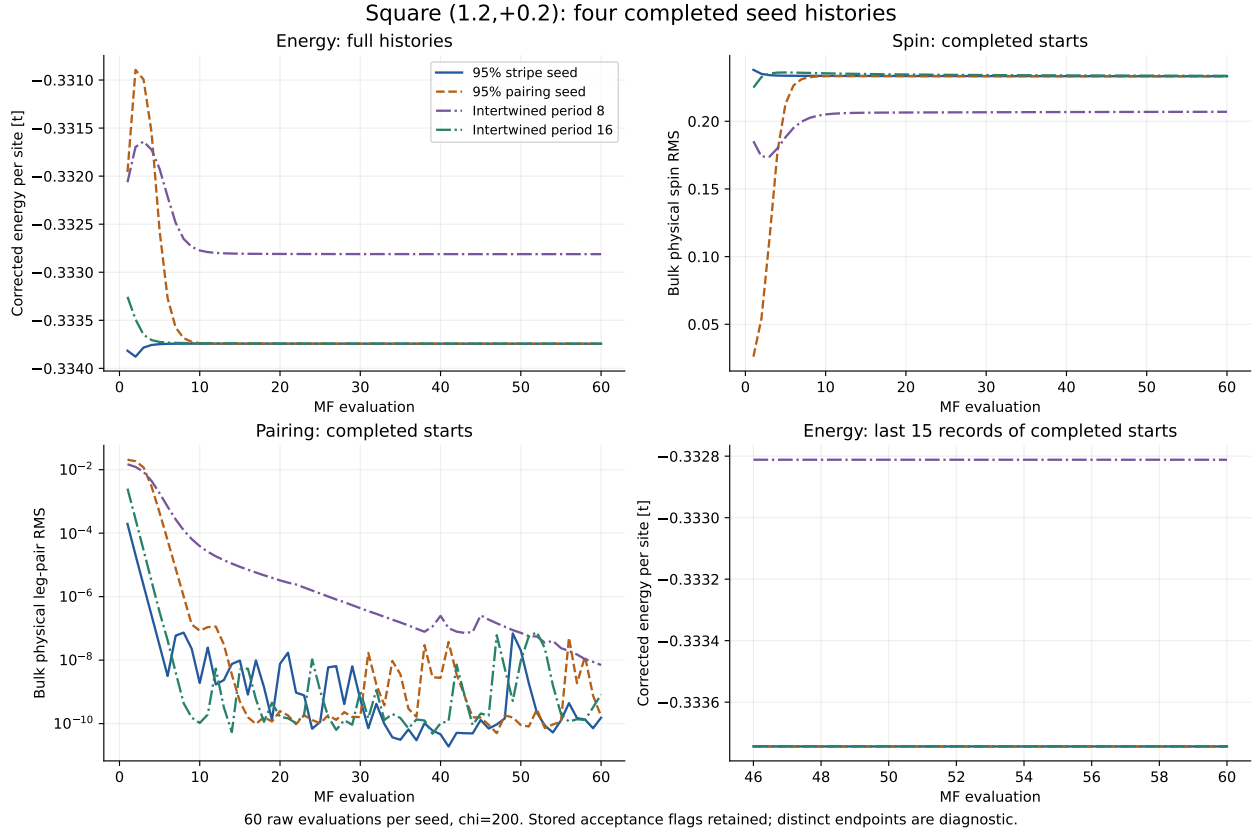


Figure 9: All four square (1.2, +0.2) histories. Full energy, spin RMS, logarithmic pairing RMS and late energy detail separate loss of pairing from persistence of different magnetic textures. Every start reaches the 60-evaluation cap without acceptance.

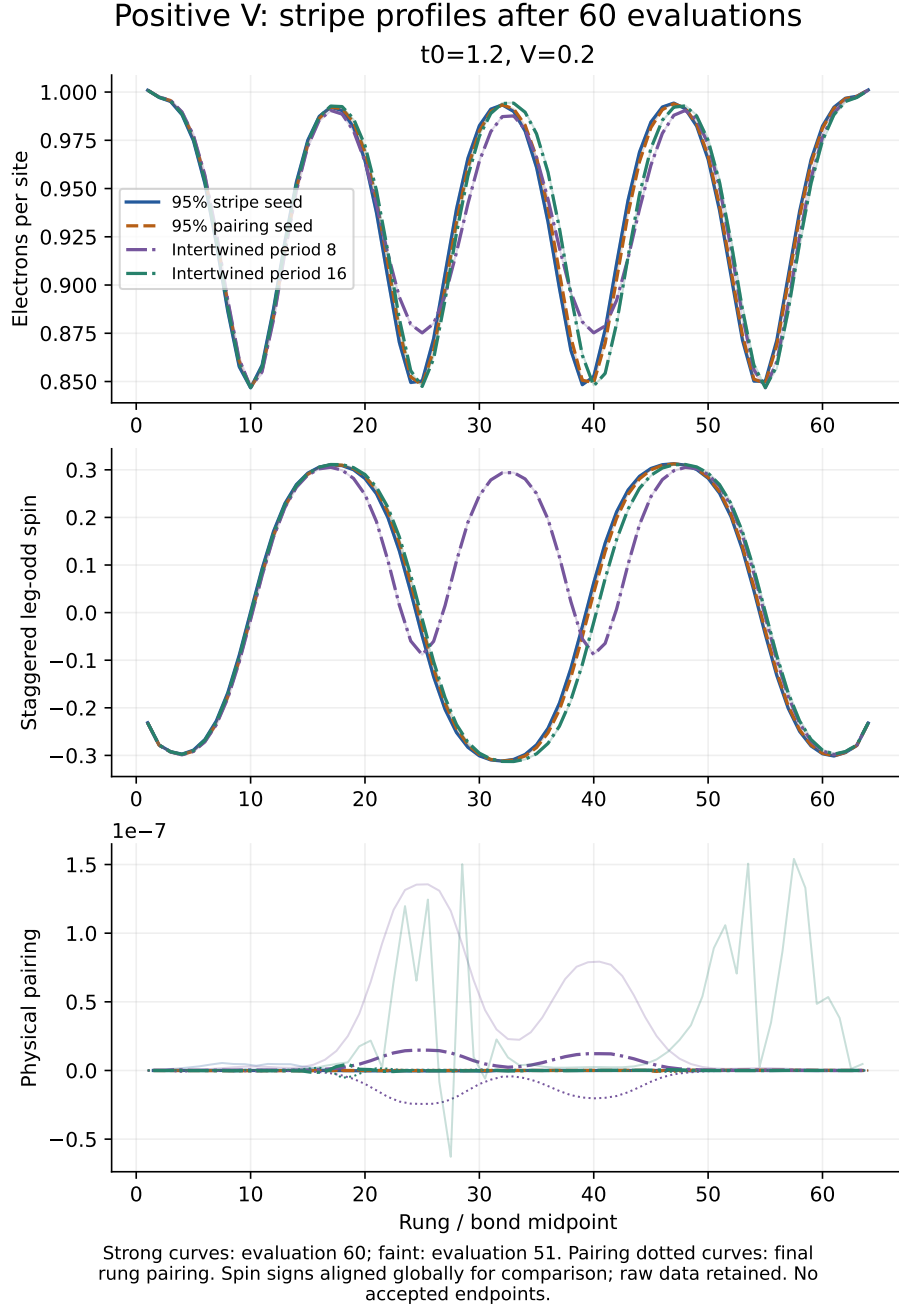


Figure 10: Square positive- V charge, staggered leg-odd spin and pairing profiles. Strong curves are evaluation 60, faint curves evaluation 51; dotted pairing curves are rung amplitudes. Only a global spin reversal aligns seeds. The period-eight start leaves two additional spin nodes. Pairing uses a 10^{-7} scale and is negligible in every endpoint; its residual spatial structure does not establish coexistence.

3.12 Trellis: spatial-cell dependence and alternating stripe relaxation

All four trellis comparisons now complete 60 raw cell sweeps at $(t_0/t, V/t) = (1, 0)$, $\tau_0/t = \tau_1/t = 0.1$, density $15/16$ and $\chi = 200$, with $L = 64$ per spatial ladder. Two seeds were tested in the skew one-ladder cell and two in the rectangular A/B cell. All four remain unaccepted maximum-iteration endpoints. The contrast in Fig. 11 is between spatial ansatzes: both one-ladder starts are pairing dominated, while both two-ladder starts develop strong, essentially unpaired stripes. The first one-ladder result must therefore be qualified by this additional cell dependence (Evidence 15).

In the one-ladder cell, the stripe seed loses spin from 0.0627 to 1.215×10^{-5} while leg-pair RMS rises from 0.00965 to 0.01796892 . The paired seed ends at spin RMS 1.368×10^{-5} and leg-pair RMS 0.01796878 . Both have mean leg/rung amplitudes near $+0.01796/ - 0.03292$: d-wave-like internal signs, not a spatial pair-density wave. Their energies, -0.518820336467 and -0.518820322856 t/site , differ by 1.361×10^{-8} t/site . Maximum seed-to-seed full-profile differences are 6.27×10^{-6} in charge and 5.68×10^{-7} in leg pairing. This supports a common paired basin, although final-ten energy ranges near 2.84×10^{-7} t/site and inner-DMRG/profile failures prevent acceptance. Small spin fluctuations persist near 10^{-5} .

Figure 12 contrasts the one-ladder paired profiles with the two-ladder stripes. Central charge standard deviation is about 0.00157 for one ladder, versus 0.0467 – 0.0470 for two. End-sensitive charge structure in the former does not establish bulk intertwining. The latter retains strong central modulation, spin RMS 0.22823 – 0.22876 on both spatial ladders, and leg-pair RMS no larger than 1.26×10^{-8} . Charge/spin bins $m = 4/30$ correspond to nominal periods $16/32$. Charge minima and spin nodes shift between A and B ; the latter is plotted at its physical offset of $-1/2$ rung. Profiles are read directly from stored correlations: trellis Hartree fields also contain normal-bond terms and cannot be inverted with the square/cubic density formula.

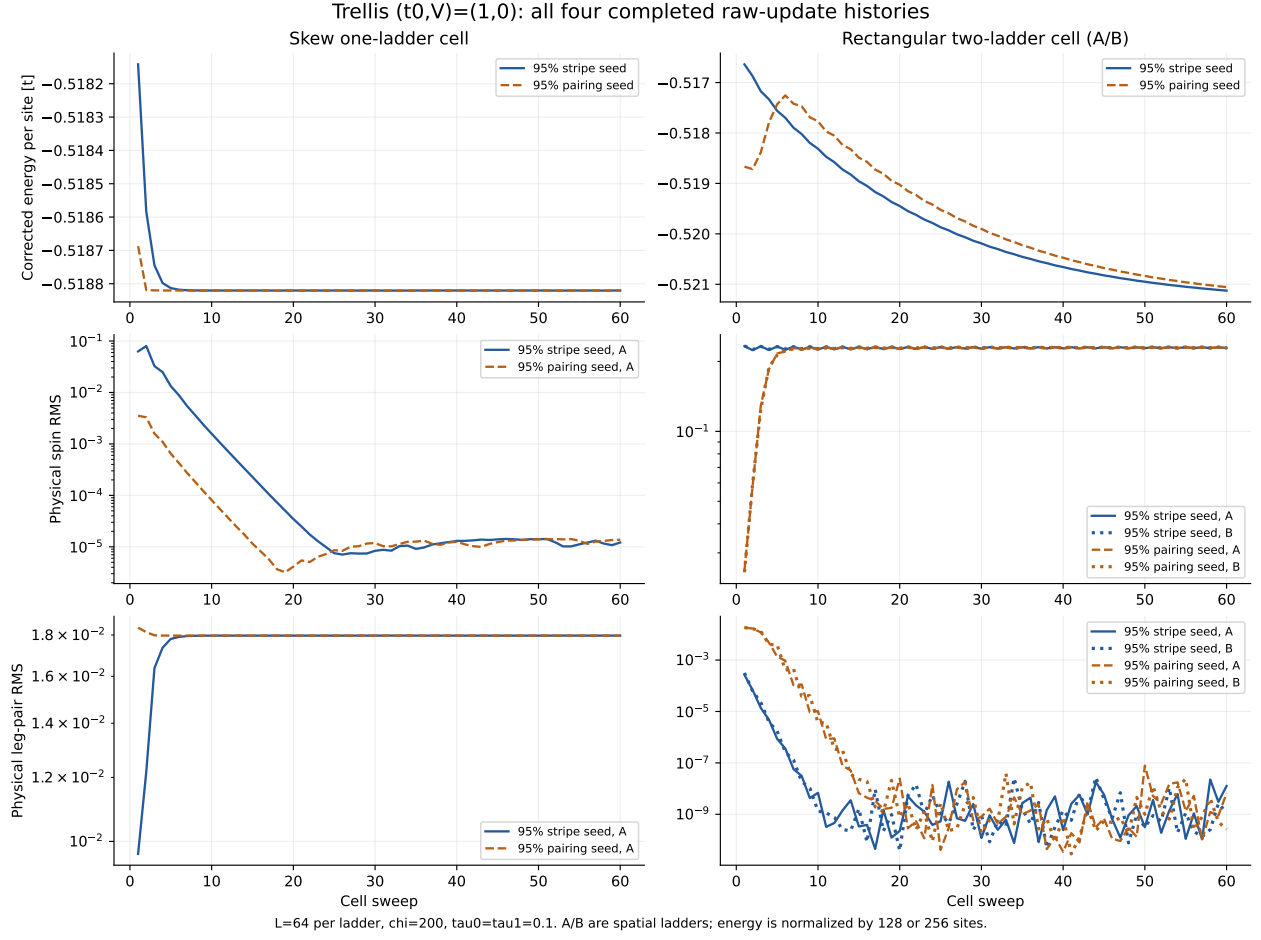


Figure 11: Complete trellis histories for both seeds in each spatial cell. Energy uses 128 sites for the skew one-ladder cell and 256 for the rectangular two-ladder cell. Spin and anomalous pairing use physical correlations and the same RMS windows as the square cuts. A/B are spatial ladders, not successive iterations or temporal orbit phases. Vertical scales differ between cells.

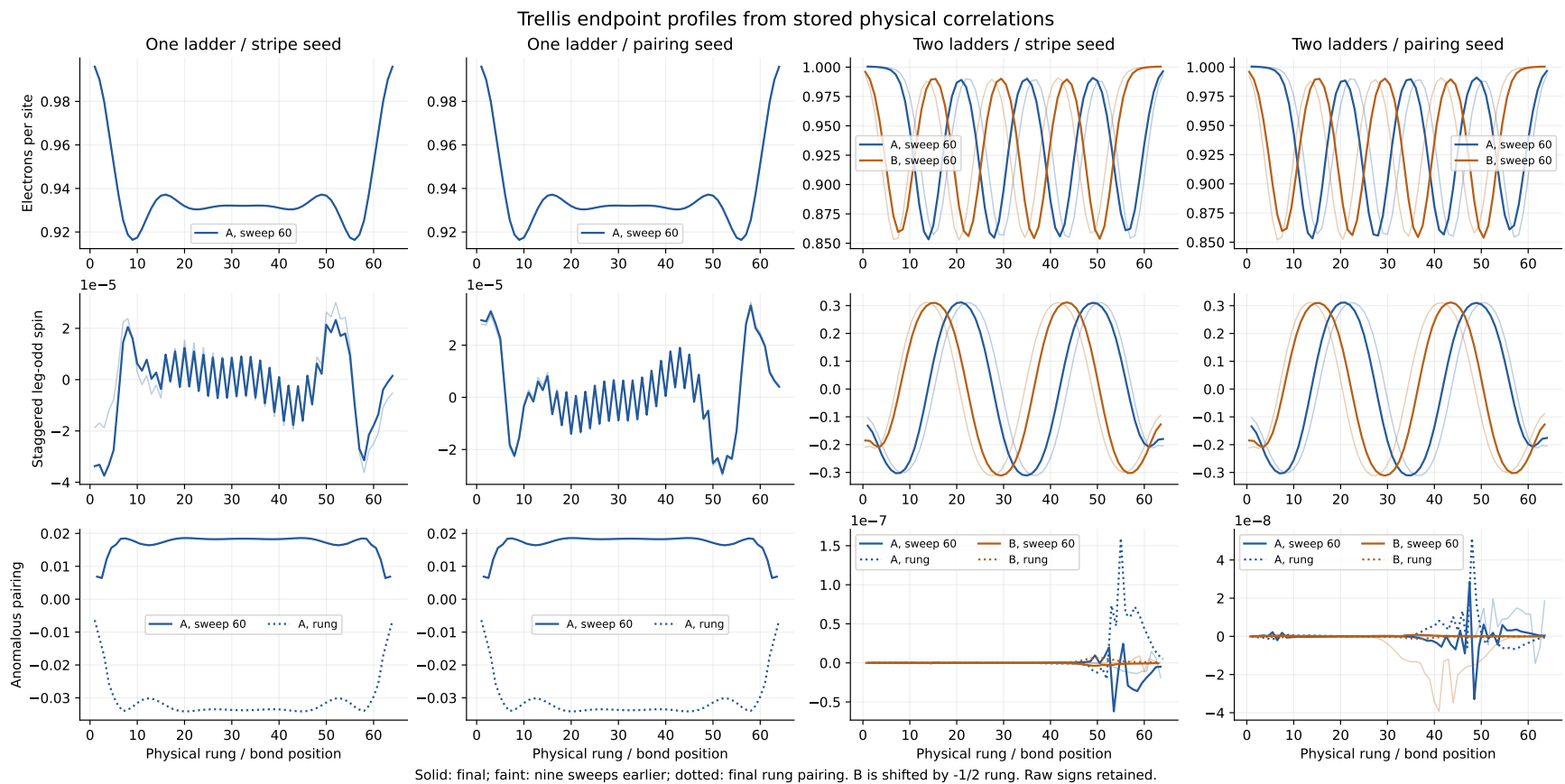


Figure 12: Trellis profiles for all four starts and six spatial ladders. Solid curves are the final sweep, faint curves nine sweeps earlier, and dotted curves final rung pairing. Raw signs are retained. One-ladder spin and two-ladder pairing panels use tiny scales; neither establishes an additional ordered component. The two-ladder spatial displacement changes with sweep parity, as resolved in Fig. 13.

Two-ladder nonacceptance is substantial, not marginal. Final global relative residuals are 0.1845–0.2085 against 10^{-4} allowed. Both final inner-DMRG windows pass, but the outer MF map and its spatial windows do not. Nearly constant spin RMS conceals the alternating wall displacement in Fig. 13. Successive full-cell measured-field increments have cosine -0.999962 and -0.999943 for the stripe and pairing starts, with norm ratios about 0.975. Maximum field changes are 0.02345/0.02620 over one sweep but 0.000754/0.000910 over two. The iteration nearly reverses its previous displacement while its amplitude slowly decreases. Two-sweep drift is still resolved. This is consistent with a damped alternating numerical mode, not an accepted period-two orbit or physical time-dependent order. Modest linear damping is a possible future diagnostic of that negative mode; no mixing change is made here.

The two-ladder stripe- and pairing-seeded endpoint energies are -0.521125816792 and $-0.521055560230 t/\text{site}$. Their difference, $7.026 \times 10^{-5} t/\text{site}$, is smaller than their summed final-ten ranges of $3.445 \times 10^{-4} t/\text{site}$. To resolve the finite-cell energy comparison, a September 20 audit places two identical copies of each saved paired ladder state in the rectangular cell and recomputes all transverse contractions there. Their target-corrected energies are -0.518817872689 and $-0.518817859078 t/\text{site}$, only $2.46378 \times 10^{-6} t/\text{site}$ above the original paired values. Both striped trials thus remain 0.002238–0.002308 t/site lower under the same finite-cell functional. Density corrections are below $8.33 \times 10^{-6} t/\text{site}$ and do not determine the ordering. The open-end embedding difference is consequently much too small to explain the observed energy advantage for these specified trials. Self-consistency is not needed to compare their expectation values. The large residuals still prevent identifying optimized stationary branches or a unique global phase minimum. Both seeds agree on the qualitative order within each cell, and a lower-energy striped trial has now been established in a common cell.

A remaining physical restriction is the transverse stripe period. After correcting the half-rung origin of B , the central charge harmonic at $q_x = \pi/8$ has a relative B -minus- A phase near 126–131 degrees. Repeating A/B alternates this phase advance and does not establish uniformly tilted, diagonal walls across many ladders. For an ideal charge profile $\delta n_m(x) = \rho \cos(q_x x + m\theta)$, a rectangular n -ladder cell requires $n\theta = 2\pi p$. Equivalently, for a rigid shift of d rungs per ladder, nd must close the longitudinal charge period and the full spin texture. Nominal charge/spin periods 16/32 with $d = 2$ would, for example, require sixteen ladders to repeat both without an additional spin transformation. This is an illustrative closure condition, not a measured tilt or proposed run size. The actual cell translation, trellis shear and leg basis must enter any such construction. Larger transverse cells or suitable translated repetitions could test oblique stripes; the current two-cell comparison does not determine their optimum.

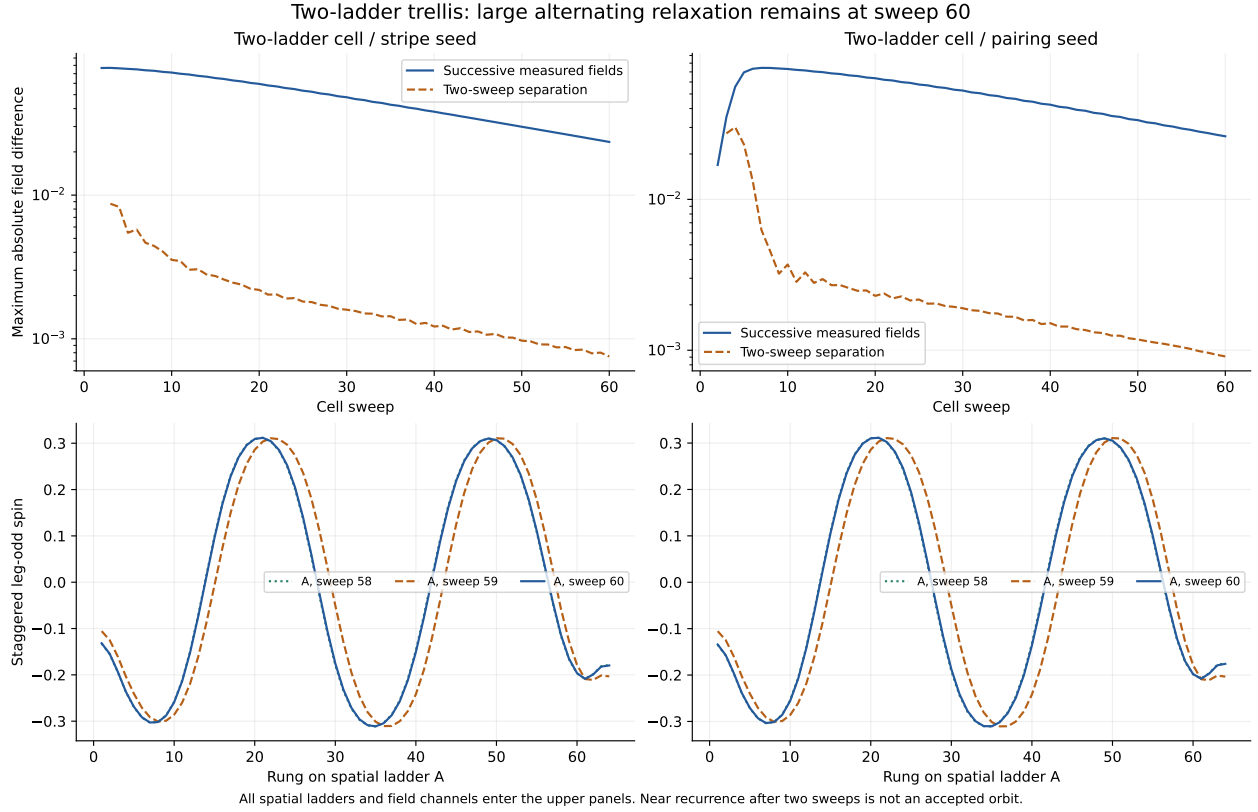


Figure 13: Two-ladder relaxation. Upper panels compare measured fields separated by one and two sweeps, taking the maximum over all channels and both spatial ladders. Lower panels resolve staggered spin on A at sweeps 58, 59 and 60. Even sweeps nearly overlap while the odd sweep has displaced nodes. Large one-step changes and continuing two-step drift preclude stationarity.

3.13 Complete pair correlations: local survival and spatial suppression

The September 22 synchronized measurements cover all 56 retrospective branches (58 spatial MPSs), plus eight diagnostics from the four newer square A/B runs. Receipts, full-source identifiers, compact-state hashes, model/configuration provenance, connected subtraction, Hermiticity and Gram positivity have been checked (Evidence 16). All retrospective branches remain unaccepted maximum-iteration snapshots. The accepted square A/B states are discussed separately below.

For this section, use the stored unnormalized singlet $D_{ab} = \sqrt{2}\Delta_{ab}$, with Δ defined in Eq. (1), and define

$$\tilde{P}_{ab,cd} = \langle D_{ab}^\dagger D_{cd} \rangle, \quad \tilde{P}_{ab,cd}^c = \tilde{P}_{ab,cd} - \langle D_{ab} \rangle^* \langle D_{cd} \rangle. \quad (15)$$

All new pair-correlation numbers and figures use this stored convention; divide by two for Eq. (1)'s normalized bond correlations. The complete basis has 318 onsite, rung and nearest-neighbor leg operators, including cross-channel entries. The older isolated reference stores only within-class Hermitian-field covariances. For disjoint bonds in that real, number-conserving state, its non-onsite field-metric factor of $1/2$ converts the sum of addition/removal correlations to $\text{Re } \tilde{P} = \text{Re } \tilde{P}^c$. Contact and overlapping-bond entries are excluded; no missing isolated rung-leg cross-correlations are inferred.

Figure 14 averages correlation magnitudes over both directions with rung bonds in 9–56 and leg-bond starts in 9–55, keeping both endpoints inside that bulk window. The isolated reference uses $\chi = 1200$, whereas coupled states use $\chi = 200$; this comparison is not a matched truncation-error study. The square stripe at $(t_0, V) = (1.4, 0)$ retains a mean connected rung magnitude 1.51×10^{-2} at separations 2–4, compared with 2.03×10^{-2} in the isolated reference. At 16–24 rungs the corresponding values are 9.81×10^{-5} and 5.22×10^{-3} . Local pairing survives, but its spatial persistence is strongly reduced in these saved states. Both stripe seeds support the same qualitative comparison.

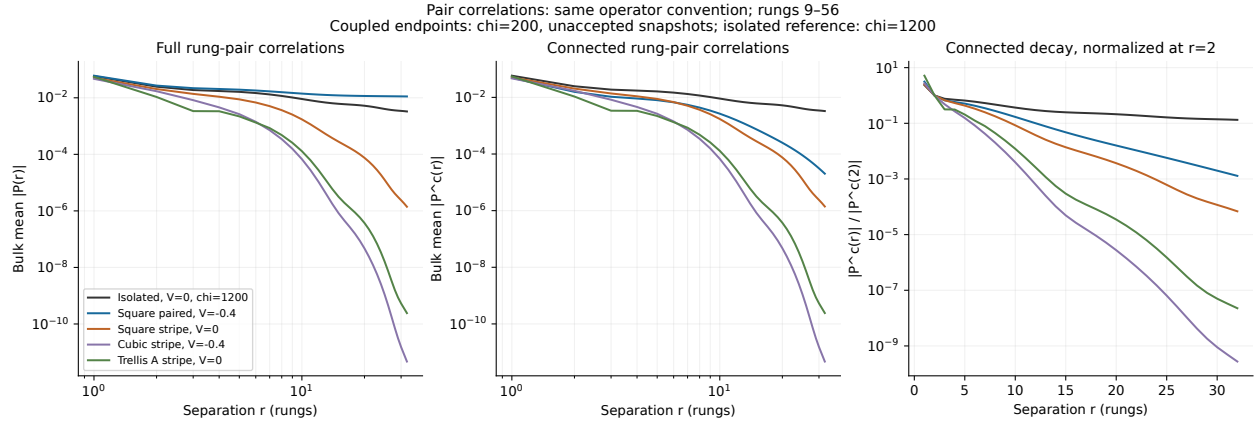


Figure 14: Full and connected rung-pair magnitudes and normalized connected decay. Figure labels P denote $\tilde{P}$ in Eq. (15). All coupled curves here are unaccepted snapshots; the isolated curve has a larger bond dimension. Square/cubic use $t_0 = 1.4$; trellis uses $t_0 = 1, V = 0$. Very small far-distance tails have no independent truncation-error estimate.

At matched intraladder parameters $(1.4, -0.4)$ and $\chi = 200$, square paired and cubic striped endpoints have comparable short-distance connected magnitudes (1.18×10^{-2} versus approximately 1.01 – 1.03×10^{-2}). At 16–24 rungs, the square value is 3.04×10^{-4} and cubic gives 1.29 – 1.39×10^{-7} , a factor of roughly 2.2 – 2.3×10^3 . The trellis comparison at $(1, 0)$ is similarly selective: short-distance

connected amplitudes in the paired one-ladder and striped two-ladder endpoints differ by only a few percent, while the long-distance value falls from 4.64×10^{-5} to approximately 6.6×10^{-7} . Figure 15 includes both seeds and spatial ladders. These are endpoint and embedding comparisons, not a causal intervention or a stationary phase ranking.

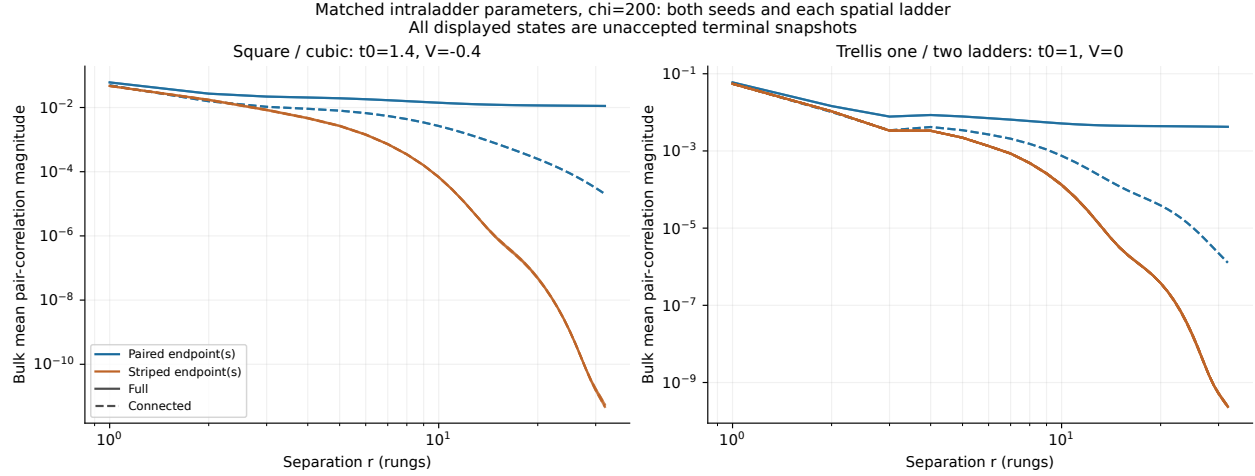


Figure 15: Matched intraladder-parameter comparisons of full and connected pair correlations at $\chi = 200$. Both initial families and each spatial ladder are shown. The states retain their original unaccepted status.

The square paired state's full long-distance rung correlation is 1.17×10^{-2} : approximately 97.4% of its signed mean is the disconnected anomalous product. Full and connected correlations instead coincide to the displayed precision in the representative stripe states. A full-correlation plateau therefore cannot be identified with slowly decaying connected fluctuations.

Relative signs also survive. In the representative square paired, square striped and cubic striped states, rung-rung and leg-leg averages are positive, while rung-leg averages are negative. Across all retrospective MPSs, the fraction of negative connected rung-leg0 entries at bulk separations 2–8 is 96.3–100% on square, 87.2–100% on cubic and 100% on trellis. This supports a surviving local rung/leg sign structure, commonly termed *d*-wave-like, without establishing a condensate or PDW. Trellis leg-leg correlations also change sign with separation, so a single uniform form factor would overstate the evidence. Signed channel curves and maps are retained in the report (Evidence 16).

The surviving short-distance pairing is spatially organized. Define $A(i) = \sum_{r=2}^4 (|\tilde{P}^c(i, i+r)| + |\tilde{P}^c(i, i-r)|)/6$ for rung-pair centers 13–52, whose six neighbors all lie inside the common bulk window. The mean A in the highest hole-density quartile exceeds the lowest quartile by factors 5.85–6.68 for square $V = 0$, 19.39–19.59 for cubic (1.4, -0.4), and 5.29–5.36 across both seeds/ladders of striped trellis. Interior hole maxima lie within half a rung of the discrete staggered-spin sign crossings (Fig. 16). These within-texture associations are descriptive; they do not establish that stripes generate an attractive interaction or a dynamical pairing mechanism.

Reference sensitivity prevents a universal decay-exponent claim. Fits of $\log |\tilde{P}^c(i, i \pm r)|$ at reference rungs 16, 24 and 32, with windows 4–12, 8–20 and 12–28 clipped to the bulk, give effective power slopes 0.39–1.36 for the isolated reference, 1.55–4.12 for square paired, -0.50 – 6.74 for square striped, 0.37–15.55 for cubic striped and 1.92–13.64 for the displayed trellis stripe. Fits require six points above 10^{-12} ; this analysis threshold is not an error bound. The negative slope is a local modulation effect, not asymptotic growth. Alternate edge cuts and same-window exponential fits

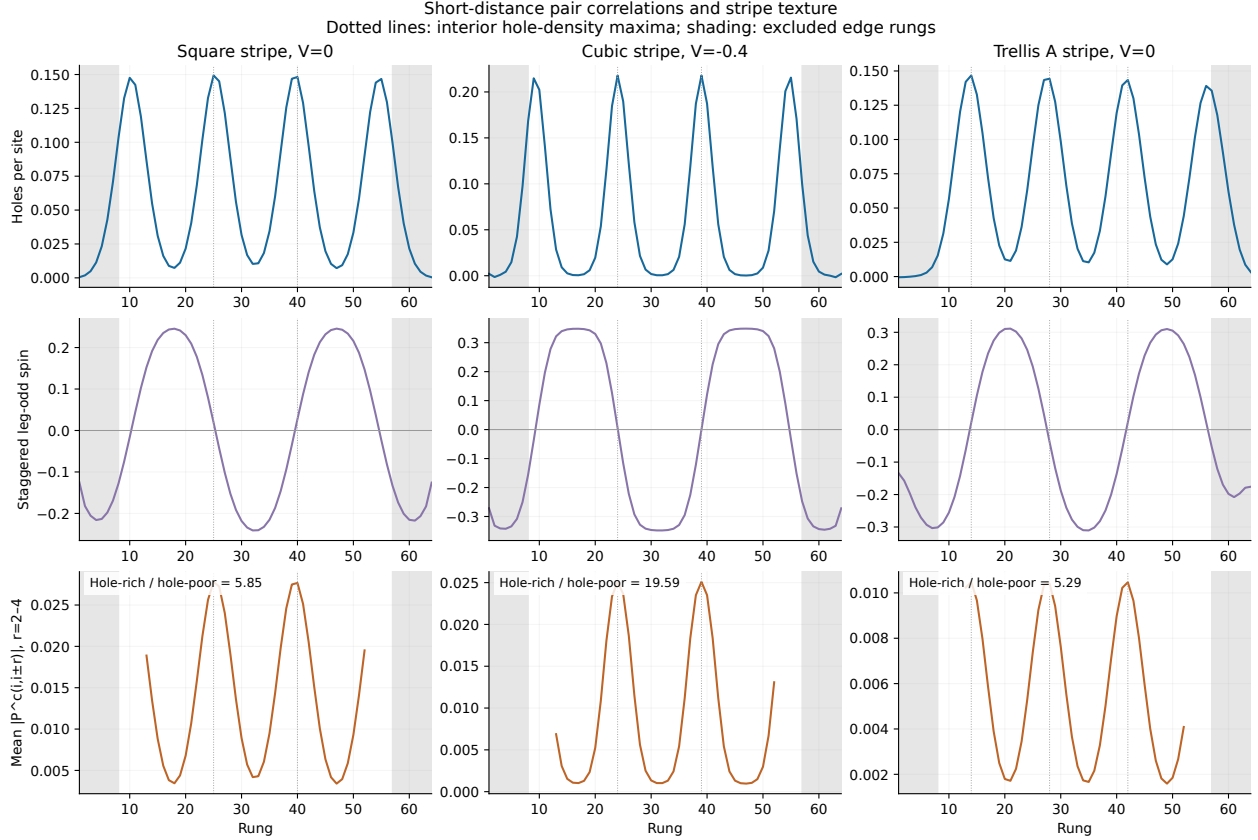


Figure 16: Short-distance pair correlations concentrate near hole-rich antiphase magnetic walls in representative snapshots. The bottom panels show $A(i)$; the middle panels show staggered leg-odd spin. Outer shading excludes eight rungs and the pair-strength centers use rungs 13–52. Dotted lines mark interior hole maxima; no independent-sample statistics or causal mechanism is inferred.

are also supplied. This sensitivity reinforces the need for reference, boundary, length and truncation controls emphasized by Shen et al. [22]. Neither the old isolated fit near 0.76 nor a selected new window should be treated as a universal exponent without those controls.

3.14 Square A/B: all four attractive runs reach paired fixed points

The square spatial extension at $t_0 = 1.4$ now has complete states and measurements for both seeds at $V = -0.4$ and -0.2 (Evidence 17). The cell contains two 64-rung MPSs with unshifted reciprocal square bonds and separate mean densities 15/16. The stripe component of ladder B's initial condition is displaced by eight rungs. Unlike the rectangular trellis result, all four square trajectories pass the fixed-point gates and lose their seeded magnetic texture (Fig. 17).

Table 3: Accepted square A/B endpoints, $t_0 = 1.4$, $\chi = 200$. The pair RMS uses the historical physical symmetrized nearest-neighbor leg anomalous expectation on bonds 6–58, not $\tilde{P}$.

V	Seed	Sweeps	Corrected $E/(Nt)$	Leg-pair RMS
−0.4	Stripe	40	−1.167404800596	0.0352251
−0.4	Pairing	40	−1.167404800545	0.0352251
−0.2	Stripe	55	−0.914117842951	0.0335678
−0.2	Pairing	44	−0.914117843004	0.0335678

The two seed energies at each V agree within $5.3 \times 10^{-11} t/\text{site}$. Bulk spin RMS is below 5.4×10^{-7} at $V = -0.4$ and 1.15×10^{-6} at $V = -0.2$. Maximum A/B leg-pair profile differences are below 1.1×10^{-6} ; charge differences reach 1.13×10^{-5} . The independent audit checks raw-update identities, source and seed hashes, channel/density/energy windows, inner-DMRG and slow-mode criteria, and reconstructs canonical energies with 256 sites per cell.

The new pair correlations also agree between seeds and ladders and reproduce the earlier paired one-ladder snapshots (Fig. 18). Connected rung magnitudes over 16–24 rungs are about 3.036×10^{-4} and 2.475×10^{-4} at $V = -0.4$ and -0.2 ; the corresponding full values are 1.166×10^{-2} and 1.030×10^{-2} . These results support survival of the paired state in this tested two-ladder square cell. They do not prove linear stability against every transverse mode, global minimality or thermodynamic superconductivity. The diagnostics remain intraladder: A/B mean-field coupling does not supply interladder quantum entanglement or a phase-stiffness measurement.

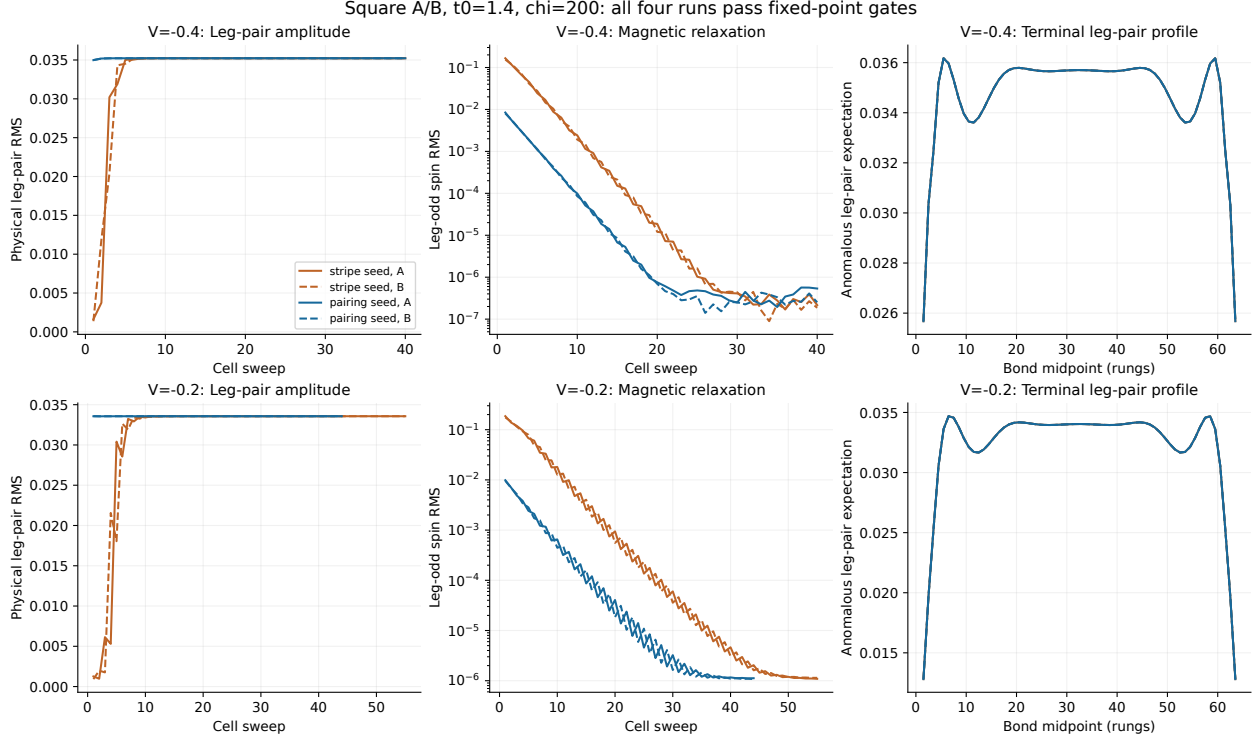


Figure 17: All four square A/B trajectories reach accepted paired fixed points. Both seed families and both spatial ladders are shown. The seed magnetization decays; the endpoint leg-pair profiles nearly coincide.

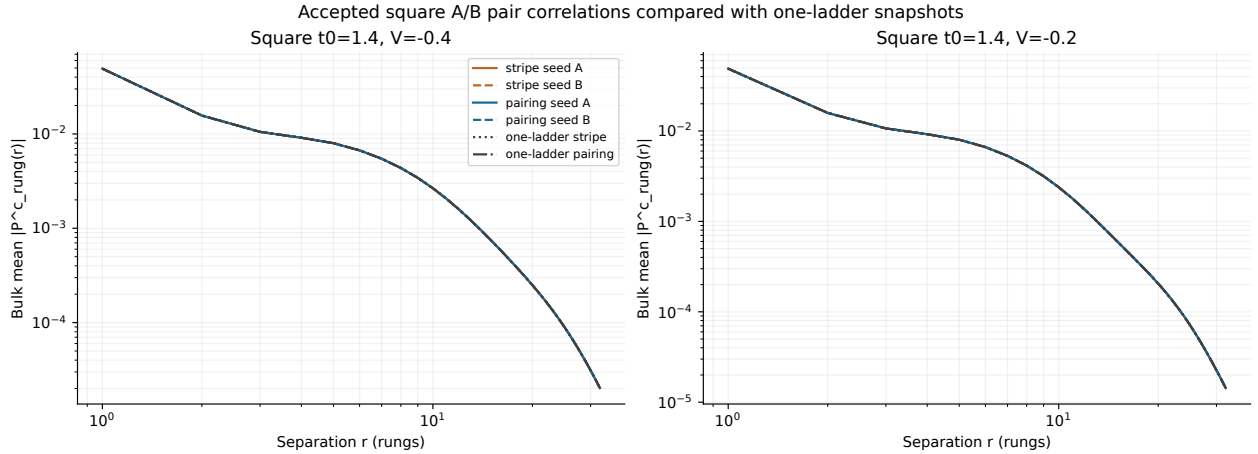


Figure 18: Connected rung-pair correlations in accepted square A/B states and earlier one-ladder snapshots. Agreement holds for both seeds and both attractions. The old one-ladder acceptance flags remain unchanged.

3.15 Pair correlations across the 42 stripe-state endpoints

The preceding comparisons establish both local pair correlations in stripe textures and uniform pairing in the tested square A/B cells. We now examine how these correlations vary across the available parameter cuts, beginning with the stripe states and then turning to the paired states in Section 3.16. The September 23 analysis uses the same 66 coupled-ladder MPSs, with $L = 64$, $\chi = 200$, $U = 8$ and target density 15/16 (Evidence 18). Seeds and spatial ladders remain separate observations, including overlaps, rather than independent statistical replicates.

Both analyses use bulk rungs $\mathcal{B} = \{9, \dots, 56\}$, $N_b = 48$, and the unnormalized rung singlet D_i from Eq. (15). To distinguish the two groups, define

$$F_i = \langle D_i \rangle, \quad F_u = \left| \frac{1}{N_b} \sum_{i \in \mathcal{B}} F_i \right|, \quad F_{\text{rms}} = \left(\frac{1}{N_b} \sum_{i \in \mathcal{B}} |F_i|^2 \right)^{1/2}. \quad (16)$$

The paired selection requires $F_{\text{rms}} > 0.01$ and $F_u^2/F_{\text{rms}}^2 \geq 0.99$. It separates a large amplitude gap: the maximum in the 42 stripe states is 1.10×10^{-4} and the minimum in the 24 paired states is 0.06579. Thresholds 0.001, 0.01 and 0.05 give the same membership, and every paired profile has at least 99.84% uniform weight. This classification uses measured pairing, rather than seed names or charge/spin amplitudes.

The stripe group comprises 20 square, 18 unfrustrated cubic and four rectangular two-ladder trellis MPSs, all unaccepted terminal snapshots. Figures 19 and 20 plot their pair correlations against the static charge and spin amplitudes. For the rung operators

$$n_+(i) = \frac{1}{2}[n_{i,0} + n_{i,1}], \quad m_-(i) = \frac{1}{2}[S_{i,0}^z - S_{i,1}^z], \quad (17)$$

the horizontal measures are

$$\rho_{\text{rms}} = \left[\frac{1}{N_b} \sum_{i \in \mathcal{B}} (\langle n_+(i) \rangle - \bar{n}_+)^2 \right]^{1/2}, \quad m_{\text{rms}} = \left[\frac{1}{N_b} \sum_{i \in \mathcal{B}} \langle m_-(i) \rangle^2 \right]^{1/2}, \quad (18)$$

where $\bar{n}_+$ is the bulk mean charge. These are static profile amplitudes. Charge RMS includes residual boundary modulation; spin RMS alone does not identify antiphase walls. The stripe interpretation also uses the spatial profiles discussed earlier. Both figures use the same 42 states and four pair measures; only the horizontal observable changes. The columns show full and connected rung-pair magnitudes from Eq. (15), each averaged over separations 2–4 or 16–24 with both rungs in the bulk and equal weight for every eligible ordered pair. The singlet is unnormalized, so these correlations are twice those of Eq. (1). Short-distance axes are linear and long-distance axes logarithmic, with common scales across geometries.

Charge texture and pair correlations. In the square row of Fig. 19, larger charge modulation broadly accompanies weaker pairing correlations, especially at longer distances. At fixed $t_0 = 1.2$, changing V from -0.4 to $+0.2$ raises charge RMS from about 0.0422 to 0.0536 for the original stripe/pairing seed families. Their short-distance connected pair magnitudes fall from about 0.0123 to 0.00745, while long-distance magnitudes fall from about 1.1×10^{-4} to 1.9×10^{-6} , roughly sixtyfold. The different fractional changes show that local pair correlations survive much more readily than their longer-distance tail.

Charge RMS nevertheless does not determine pairing by itself. At $V = 0$, the square $t_0 = 1.2$ and 1.4 endpoints have nearly the same charge RMS, about 0.052, but the latter have larger

short-distance pair correlations (0.0141–0.0151 versus 0.0090) and roughly 11–13 times larger long-distance correlations. In the cubic row, charge RMS is larger (0.0795–0.0899), short-distance connected pairing remains 0.00317–0.01034, and long-distance values span approximately 1.9×10^{-12} – 1.4×10^{-7} . This geometry comparison illustrates strong suppression of extended pairing despite finite local pair correlations, but changes the coupling geometry as well as the charge texture. The smallest tails have no independent error bound.

Magnetic texture and pair correlations. Figure 20 makes the broad anticorrelation with static magnetic amplitude particularly clear. Square spin RMS ranges from about 0.149 to 0.262. The weakly magnetic stripe endpoints near the paired boundary retain short-distance pair magnitudes up to 0.0162 and long-distance values around 3×10^{-4} . More strongly magnetic square endpoints have short-distance magnitudes around 0.005–0.006 and tails around 10^{-7} – 10^{-6} . Along the $t_0 = 1.2$ cut above, spin RMS rises from about 0.184–0.185 to 0.233 as both pair measures decline. Cubic endpoints extend to still larger spin RMS, 0.280–0.354, with much smaller pair tails. These are descriptive trends across parameters and geometries, not a universal scaling law.

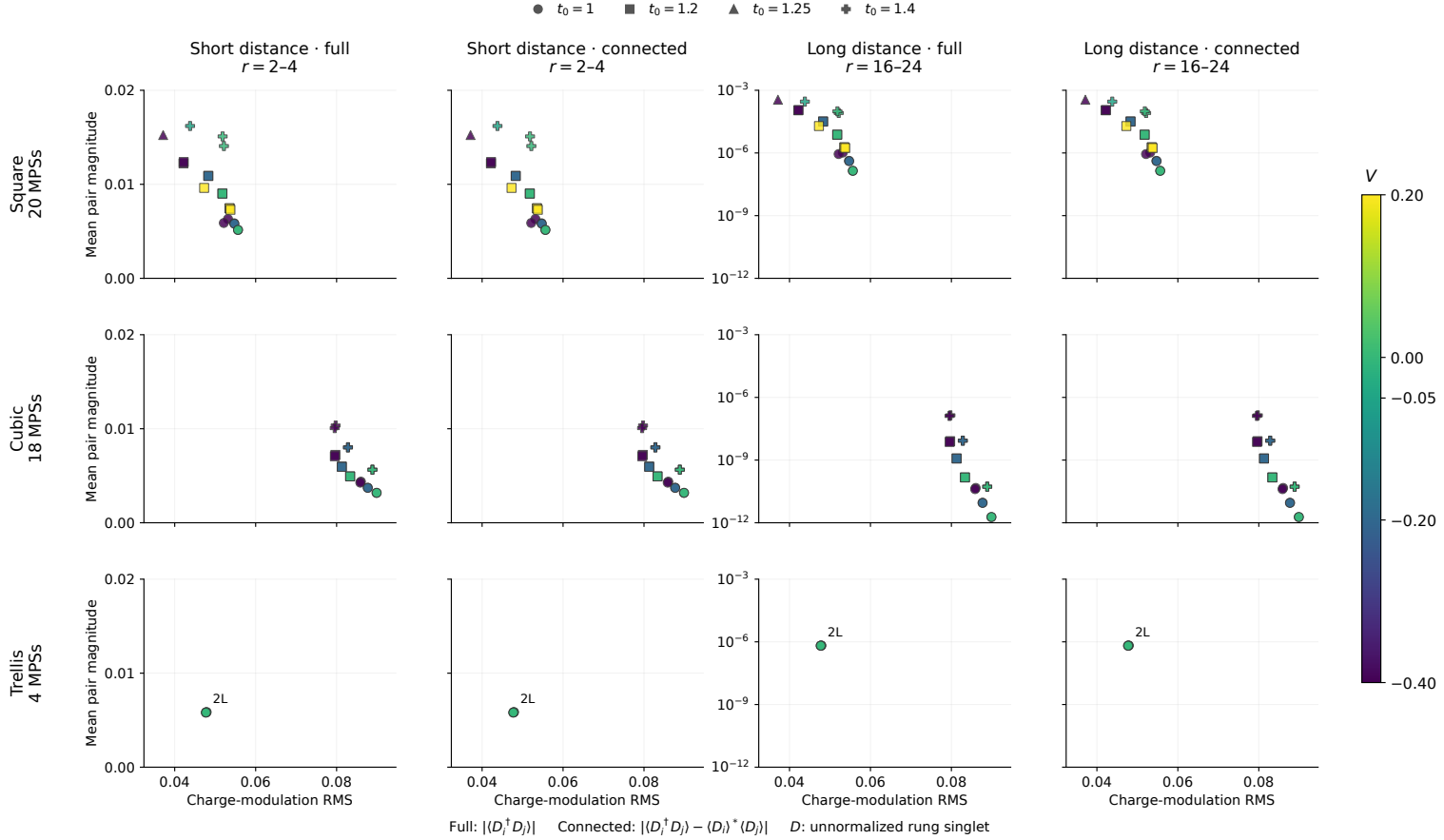
The four square starts at $(t_0, V) = (1.2, +0.2)$ also demonstrate why seed outcomes remain separate. The period-eight start retains weaker static amplitudes ($\rho_{\text{rms}} = 0.0473$, $m_{\text{rms}} = 0.2038$) and a larger long-distance pair magnitude (1.94×10^{-5}) than the other three starts ($\rho_{\text{rms}} \simeq 0.0536$, $m_{\text{rms}} \simeq 0.233$, pair magnitude 1.7 – 1.9×10^{-6}). This is a comparison of unaccepted textures at the same parameters, consistent with the previously identified defect/coarsening remnant; it does not establish a stable intertwined phase.

Full versus connected correlations and the trellis subset. The full and connected columns nearly coincide because anomalous pairing is tiny in this group. Their largest relative difference is below 8×10^{-7} at short distances and 4.3×10^{-5} at long distances. The observed pair correlations therefore survive the removal of the anomalous product; they are not a disconnected plateau from uniform pairing. The four trellis MPSs represent A/B ladders from two seeds at one parameter coordinate. They cluster near charge RMS 0.0478 and spin RMS 0.229, with short-distance connected pairing about 0.00584 and long-distance pairing about 6.6×10^{-7} . They establish agreement between these saved textures, not a trellis parameter trend or four independent realizations.

Taken together, the 42-state grids support coexistence of static stripe textures with finite local pair correlations, while stronger static textures broadly accompany poorer spatial persistence of pairing. These parameter associations describe unaccepted stripe textures, so they do not establish that stripes cause or are necessary for pairing. The complementary question is whether charge and spin correlations persist after uniform anomalous pairing develops, and how they vary with its strength. We address that question next within the paired square subset.

Stripe states: stripe amplitude versus pair correlations

Uniformly paired states removed. Each mark is one spatial MPS; overlapping seeds/A/B ladders remain separate.



Common bulk: rungs 9–56; $L=64$, $\chi=200$, $U=8$, target $n=15/16$. Long-distance axes are logarithmic.

42 stripe-state endpoints, all unaccepted. Parameter associations; tiny tails have no independent error bound.

Figure 19: Pair correlations versus static charge RMS in the 42 stripe-state MPSs, with all uniformly paired states removed. Rows separate square, cubic and trellis geometries; columns separate short/long distances and full/connected correlations. Color denotes V and marker shape t_0 . Definitions and analysis are in Section 3.15. All endpoints remain unaccepted; overlapping seeds and A/B ladders are retained.

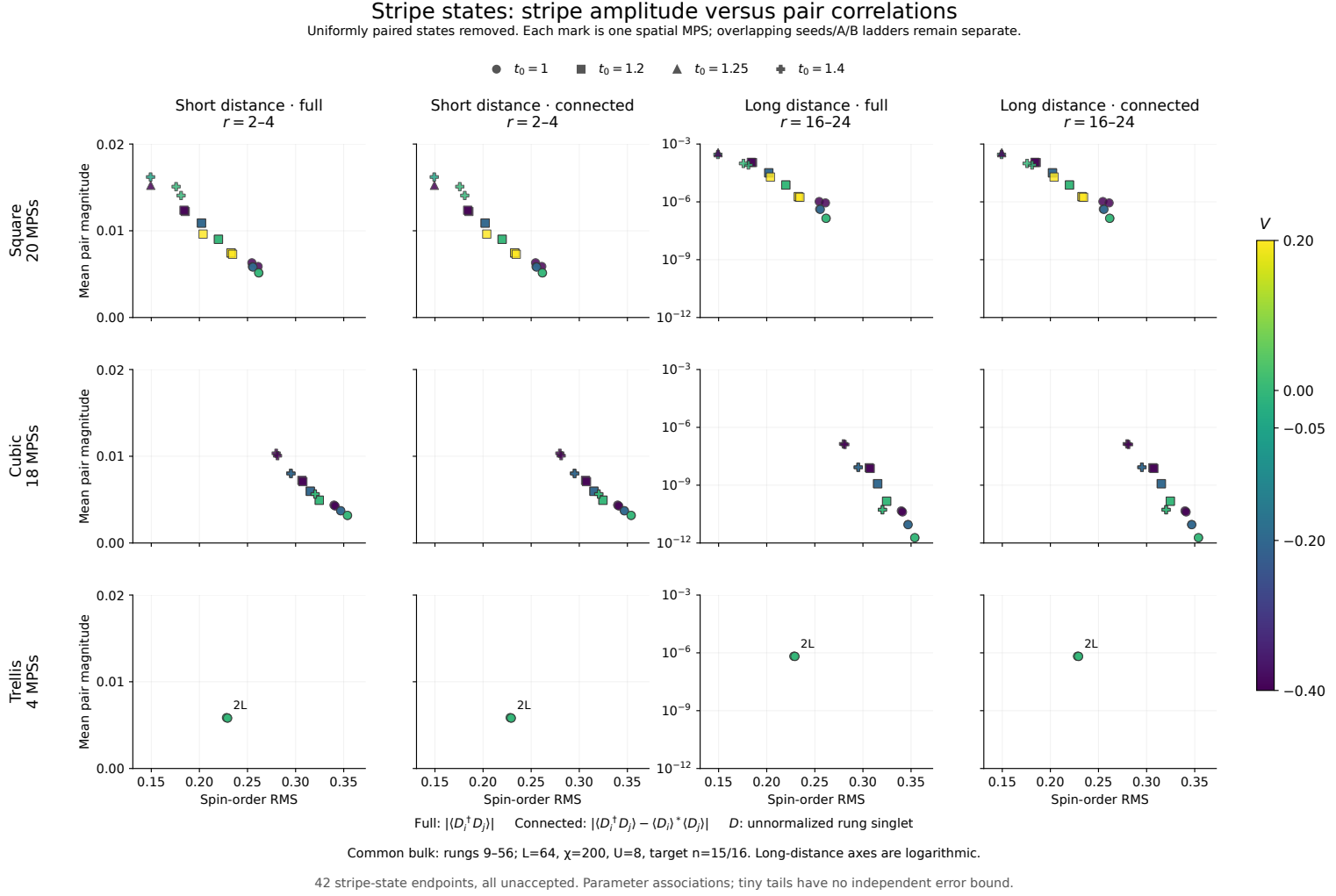


Figure 20: The same 42 stripe-state MPSs and pair measurements versus static spin RMS, with the layout and conventions of Fig. 19. The broad suppression of pairing with increasing magnetic amplitude is stronger at long distances. Static RMS, connected pair correlations and anomalous pairing are distinct observables; no acceptance status or phase ranking is changed by this comparison.

3.16 Stripe correlations within uniformly paired square states

We now reverse the comparison: charge and spin correlations are measured against the uniform pair amplitude F_u defined in Eq. (16). Figures 21 and 22 show 22 paired square MPSs: eight from accepted A/B fixed points and fourteen unaccepted snapshots. The two paired trellis MPSs share one parameter coordinate and remain only in the source tables. Here F_u is a physical anomalous expectation, distinct from the MF source α and the leg-pair RMS in Table 3. Dividing it by $\sqrt{2}$ gives the normalized rung-singlet amplitude of Eq. (1).

Using the charge and spin operators in Eq. (17), we remove the static one-point contribution and form the connected covariance and its Fourier sum:

$$C_O(i, j) = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle, \quad S_O^c(q) = \frac{1}{N_b} \sum_{i, j \in \mathcal{B}} e^{iq(i-j)} C_O(i, j). \quad (19)$$

Figure 21 evaluates the leg-even charge channel at $q_c = \pi/8$ and the leg-odd longitudinal-spin channel at $q_s = 15\pi/16$, referencing the previously identified charge period 16 and spin-envelope period 32. These are fixed-wavevector weights, not fitted peak heights. Their normalization uses rung averages and N_b ; it differs from a site-normalized structure factor. Contact terms $i = j$ are included, so a finite weight can contain local variance and broad short-range correlations without a resolved stripe peak.

The square trend depends on the parameter cut. At fixed $V = -0.4$, increasing t_0 from 1.25 to 1.4 increases F_u from 0.09344 to 0.10637, while $S_{m-}^c(q_s)$ falls from 0.35611 to 0.32420, approximately 9%. At fixed $t_0 = 1.4$, making V more attractive from -0.05 to -0.4 also strengthens pairing (0.09709 to 0.10637), but the spin weight instead rises from 0.31331 to 0.32420, approximately 3.5%. These statements describe the paired branch; near the boundary the other seed can remain striped and is excluded from this subset. Across the square paired points the charge weight occupies the narrow interval 0.04637–0.04745, with some nonmonotonicity. The scatter therefore does not support a single parameter-independent relation between pairing strength and stripe-channel correlation weight.

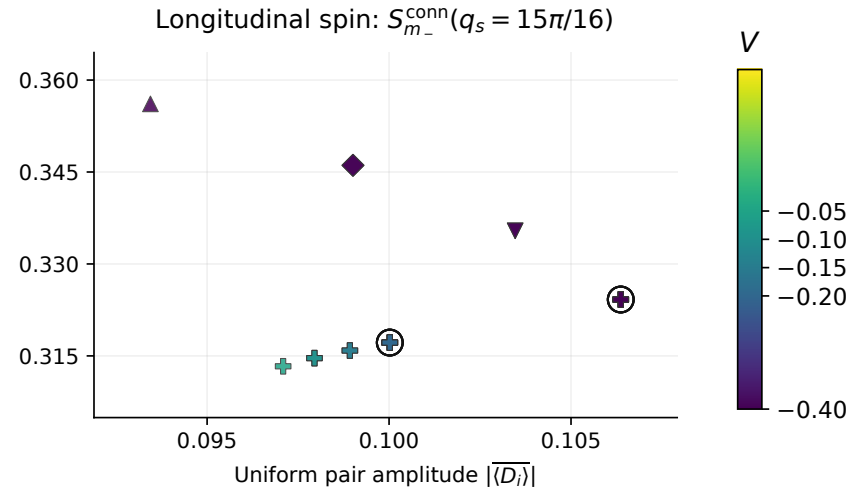
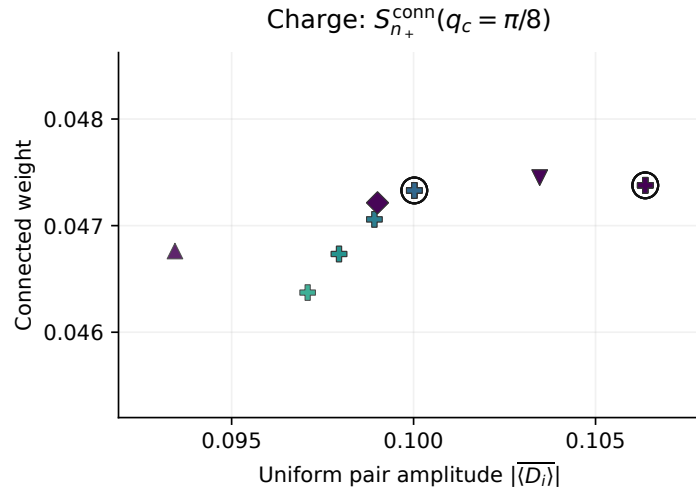
The distance grid separates local weight from spatial range. Figure 22 shows the mean of $|C_O(i, j)|$ over $2 \leq |i - j| \leq 4$ and $16 \leq |i - j| \leq 24$, with both endpoints in $\mathcal{B}$ and equal weight for each eligible ordered pair. These measures exclude contact terms but do not select a wavevector. Along the fixed- V hopping cut above, the long-distance spin magnitude drops from 4.44×10^{-6} to 4.56×10^{-7} , nearly a factor of ten, whereas the short-distance magnitude falls from 0.01502 to 0.01053. The long-distance charge magnitude also decreases, from 2.64×10^{-5} to 1.68×10^{-5} . Along the fixed- t_0 interaction cut, the spin magnitudes instead increase: the short-distance value rises from 0.00976 to 0.01053 and the long-distance value from 3.50×10^{-7} to 4.56×10^{-7} . Long-distance charge likewise rises from 1.26×10^{-5} to 1.68×10^{-5} ; short-distance charge is nonmonotonic. The finite- q and distance measures therefore give complementary information. In particular, sizable spin weight at q_s does not imply a large long-distance spin tail. The smallest tails remain descriptive finite- χ values without independent error bounds.

Thus uniform pairing coexists with finite connected charge/spin correlations, whose dependence on pairing varies between parameter cuts. This complements local pair survival in the stripe states. Because V and t_0 change both channels, these associations establish neither simultaneous static stripe order nor a causal requirement for stripes.

Uniformly paired square states: stripe correlations versus pairing strength

22 square MPSs; overlapping seeds and A/B ladders remain separate points.

▲ $t_0 = 1.25$ ◆ $t_0 = 1.3$ ▼ $t_0 = 1.35$ + $t_0 = 1.4$ ○ Outer ring: accepted



$$S_O^{\text{conn}}(q) = N_b^{-1} \sum_{ij} e^{iq(i-j)} [\langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle]; \text{ fixed stripe wavevectors, contact terms included.}$$

$$n_+ = (n_0 + n_1)/2, m_- = (S_0^z - S_1^z)/2; \text{ bulk rungs 9-56. Axis ranges differ by measure.}$$

14 unaccepted + 8 accepted MPSs. Equal-time intraladder correlations; parameter associations, not causal response functions.

Figure 21: Connected charge and longitudinal-spin weights at the fixed stripe wavevectors versus uniform rung-pair amplitude in the 22 paired square MPSs. Operators, bulk window and normalization are defined in Eqs. (16), (17) and (19). Color indicates V , shape indicates t_0 , and outer rings identify the eight accepted MPSs. Axis ranges differ by panel. Contact terms are included; these values do not establish stripe peaks. Overlapping seeds and A/B ladders remain separate observations.

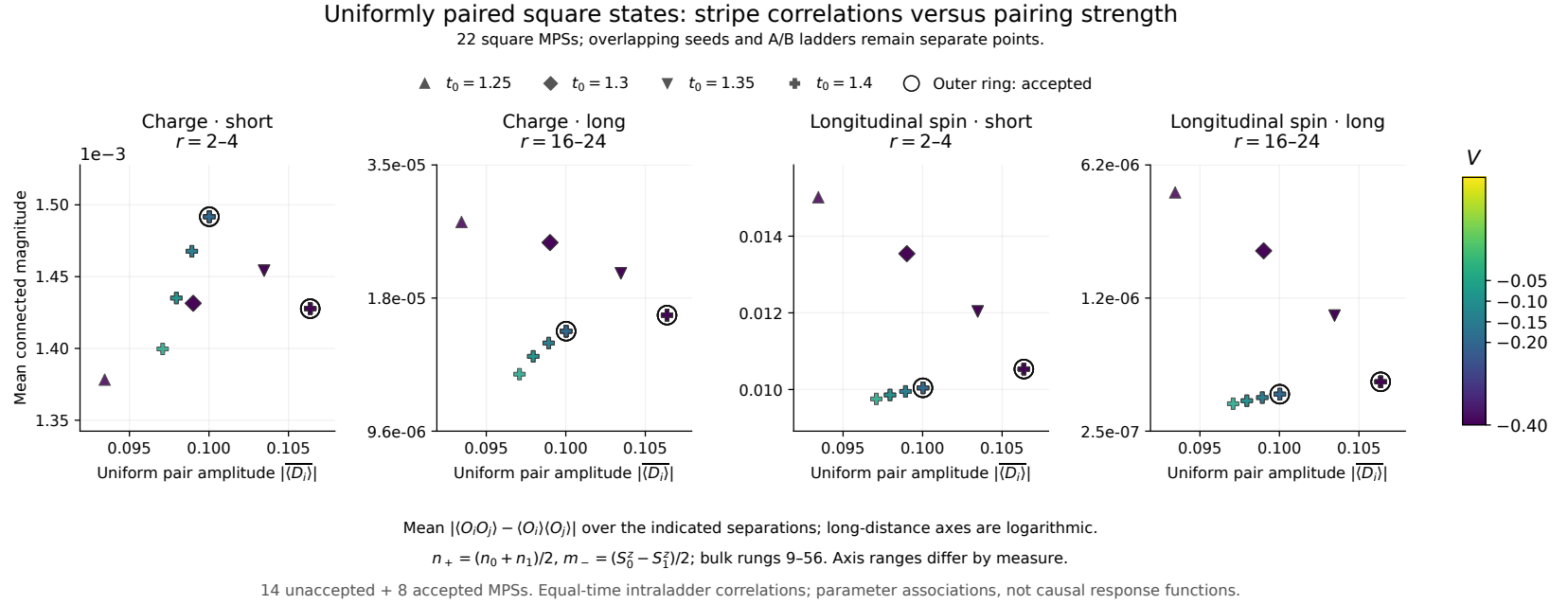


Figure 22: Short- and long-distance connected charge and longitudinal-spin magnitudes versus uniform pairing in the same 22 square MPSs. Both rungs lie in 9–56; each panel averages $|C_O(i, j)|$ over its indicated separation window. Contact terms are excluded, and the long-distance axes are logarithmic. These spatial measures are not restricted to a stripe wavevector. Marker conventions match Fig. 21; the opposite trends along the two square parameter cuts remain visible.

3.17 Scope of the emerging physical interpretation

The isolated-ladder results, array comparisons and correlation grids together separate local pairing tendencies from ordered-state selection. A spin-gapped ladder with slowly decaying pair correlations can still feed a static stripe state once transverse normal fields are allowed to develop. At the attractive anchor, both tested starts instead approach the same pairing-dominated solution. The current evidence therefore supports a model-specific competition between collective orders within a system that already possesses local pairing tendencies.

This result shares an ingredient with stripe-centered pairing mechanisms: correlated low-dimensional subsystems can support pairing before collective coherence is established. At the level of the current static textures, however, the anchor comparison resembles competition between stripe order and superconducting order. It neither establishes that fluctuating stripes are necessary for pairing nor excludes their possible beneficial role. The approximation treats quantum fluctuations within the ladder, but does not resolve the full collective stripe and phase fluctuations of a plane.

The completed four-fermion measurements now resolve part of the earlier open question: the measured stripe endpoints retain local singlet-pair correlations near their hole-rich walls, while longer-distance correlations are strongly suppressed. This supports a distinction between local pairing and spatially extended order. The stripe-state grids show that stronger static textures broadly accompany weaker pair tails, while the paired-state grids show that residual magnetic correlations can increase or decrease with pairing along different parameter cuts. These associations do not isolate phase fluctuations as the cause, establish fluctuating superconductivity or certify the unaccepted stripe endpoints as stationary phases. The newly accepted square A/B paired states strengthen the paired side of the comparison within that specific spatial extension.

Attraction changes more than one ingredient of the coupled problem. The prefactor $g = t_{\perp}^2/|E_p|$ decreases when $|E_p|$ increases at fixed $t_{\perp}$, even while the ladder's pair response can become stronger. This motivates separating the change in the bare ladder from the change in the coupling kernel across a V scan. Recent MPS+MF work on a different, mixed-dimensional t - J model likewise emphasizes the properties of the one-dimensional building blocks and the requirements for an ordered array [62]. That work is retained as an August 2026 preprint, not a result for the present Hamiltonian.

3.18 Current limitation: equal ladder densities and parallel stripes

The excluded charge mode. The present spatial extension gives ladders A and B independent profiles, but targets the same average electron density on each ladder separately. Writing x for the longitudinal rung coordinate and m for the ladder index, the implemented constraint is

$$\bar{n}_m = \frac{1}{2L} \sum_{x=1}^L \sum_{\ell=0}^1 \langle n_{m,x,\ell} \rangle = n_{\star} \quad \text{for every } m. \quad (20)$$

This is stronger than fixing only the total density of the spatial cell. For example, it excludes $\bar{n}_A = n_{\star} + \Delta n$ and $\bar{n}_B = n_{\star} - \Delta n$ for nonzero Δn . Hole-rich and hole-poor ladders could then form stripes running parallel to the ladder direction, with charge modulated across the array. In wavevector language, the excluded ladder-average component has $q_{\parallel} = 0$ and nonzero $q_{\perp}$. Density waves varying along each ladder can still develop, including different longitudinal phases on A and B. The restriction does not forbid every transverse texture: redistribution between the two legs can

preserve a ladder’s average density. It specifically removes interladder charge disproportionation, irrespective of how accurately the allowed states converge.

What ladder experiments establish. Resonant X-ray scattering in $\text{Sr}_{14}\text{Cu}_{24}\text{O}_{41}$ found a hole crystal with a period of about five rungs *along* the ladders, coherent across roughly 50 neighboring ladders [27]. Ca-dependent measurements found longitudinal periods of five and three, with a strong commensurability dependence [28]. These results demonstrate electronic charge ordering in a coupled-ladder material. Their transverse coherence does not establish different average densities on neighboring ladders: a common longitudinal density wave can be coherent across an array with equal ladder averages. The cited experiments therefore support an ordering channel permitted by Eq. (20), and do not directly demonstrate the omitted parallel-stripe state. Conversely, they are not an exclusion test for that state. The chain reservoirs, multiorbital interactions and structural geometry also prevent a direct identification of the material with the present one-band model.

Spontaneous charge order in numerical ladder studies. The four-leg t - J work of White and Scalapino establishes stripe states without an imposed bulk charge potential [63]. Their six-leg Hubbard calculation finds stripes wrapped around the cylinder, so the density varies along its length [64]. Such finite-width results are evidence for spontaneous stripes, but do not by themselves compare both orientations in a broad array of weakly coupled two-leg ladders. Boundary pinning and the available transverse wavelengths must be distinguished from an imposed bulk modulation. Furthermore, pronounced density oscillations in an isolated open ladder need not imply long-range CDW order; the Friedel-oscillation analysis of White, Affleck and Scalapino supplies the relevant distinction [61].

A particularly close methodological precedent is the DMRG plus interladder mean-field study of Almeida, Roux and Poilblanc [65]. Uniform t - J ladders coupled by screened Coulomb repulsion develop longitudinal charge modulations with periods three, four and five. Trellis frustration affects their relative registration; the period-four solution involves two inequivalent ladder profiles. However, the calculation uses specified candidate CDW patterns around a common mean density and does not establish stability against arbitrary interladder charge transfer. Its “pair density waves” are crystals of hole pairs without superconducting coherence, rather than evidence for a sign-changing superconducting pair field. This work supports spontaneous longitudinal charge ordering within an approximation related to ours, without resolving the omitted channel.

Why parallel stripes remain a credible competitor. Two-dimensional orientation studies provide a stronger motivation to release the constraint. In an anisotropic t - J model, Kampf, Scalapino and White find that exchange anisotropy favors stripes along the stronger-exchange direction in the regime studied; their estimate uses a boundary-pinned cylinder [66]. The iPEPS calculations of Corboz, Rice and Troyer compare stripe orientations in the thermodynamic limit. For $t_x = 0.85t_y$, $J_x = 0.85^2J_y$ and $J_y/t_y = 0.4$, low-doping stripes prefer the stronger-coupling direction, whereas the preferred orientation reverses at higher doping, above approximately $\delta = 0.14$ in that calculation [67]. The directional anisotropy is spatially uniform; a charge modulation is not built into the Hamiltonian. Thus, neither orientation follows from a universal kinetic-energy argument. The multi-method evidence for stripes in the uniform square-lattice Hubbard model also establishes a substantial competing family of inhomogeneous states [11]. In the isotropic infinite square lattice, bond-aligned orientations are related by rotation. Our physical ladder anisotropy and trellis frustration remove that equivalence, so these results motivate a test but do not predict

its outcome. Models with deliberately imposed bulk stripe modulation, including Ref. [15], are not used as evidence that the missing density pattern forms spontaneously.

Why the restriction may be benign in part of the phase diagram. An energetic argument separates a possible instability from its necessity. For two identical decoupled ladders with a smooth energy per site $e(n)$, the redistribution cost per paired set of sites is

$$e(n_\star + \Delta n) + e(n_\star - \Delta n) - 2e(n_\star) = e''(n_\star)(\Delta n)^2 + O((\Delta n)^4). \quad (21)$$

Positive inverse compressibility $e''(n_\star) > 0$ penalizes this mode. Interladder energy gains and any accompanying spin reconstruction must overcome that cost. Strong finite-wavevector correlations along a ladder, or a spin gap, do not by themselves establish an instability of its average density. This is a conditional argument, not a measured curvature or stability result for our endpoints; near phase separation, a charge plateau, or a first-order redistribution, the smooth small-amplitude expansion is insufficient. The available ladder experiments and coupled-ladder CDW calculations therefore make the retained channel physically reasonable, while leaving the quantitative importance of the excluded channel open.

Interpretation boundary and a prospective test. The current conclusions compare states within the equal-average-density ladder sector. In particular, convergence of both square A/B seeds to the paired state establishes reproducibility within that sector, not stability against parallel stripes. A first extension would use one common external chemical potential to impose $(\bar{n}_A + \bar{n}_B)/2 = n_\star$, allow independent ladder densities, and test small imbalance seeds. The common chemical potential would coexist with different self-consistent local Hartree fields. The density gate, canonical functional and effective-coupling assumptions would have to remain consistent with the released constraint. Small-seed relaxation tests only local stability; a competing finite-amplitude minimum would require additional seeds or constrained energy comparisons.

Two ladders would still allow only a short transverse repeat. For a square array, if the excess holes are organized into stripes with linear filling ρ_ℓ holes per longitudinal lattice spacing, counting gives $p = \rho_\ell/\lambda_\perp$, where $\lambda_\perp$ is the spacing in physical rows. At our $p = 1/16$, illustrative half-filled and filled stripes would have spacings of eight and sixteen rows, respectively, corresponding to four and eight two-leg ladders. These are conditional examples, not selected fillings; antiphase spin order can require a doubled repeat. Consequently, absence of imbalance in an A/B cell would not rule out longer-period parallel stripes. No charge-transfer stability calculation or larger-cell campaign is included in the present evidence.

3.19 Controls needed before stronger claims

The pair-breaking expansion needs a careful energy-scale comparison. The backbone gives $t_\perp/|E_p| \simeq 0.683$ and $t_\perp/\Delta_s \simeq 0.652$ at $t_\perp/t = 0.1$; these ratios are not parametrically small. The quantity

$$\Delta_{c,2} = \frac{E(N+2) + E(N-2) - 2E(N)}{2} \quad (22)$$

instead measures even-particle charge curvature. A gapless charge mode can coexist with a finite spin or pair-breaking scale, so $t_\perp/\Delta_{c,2} > 1$ does not by itself invalidate the projection. This corrects the stronger interpretation in the earliest bare-ladder report (Evidence 5). The registry and backbone also use slightly different binding energies, differing by about 0.0188% in the inferred coupling; quantitative response/SCF comparisons should declare a common kernel.

The remaining high-value controls are:

1. Complete self-consistency at the representative stripe point and compare matched canonical energies only where the states and provenance support it. A converged fixed point and a global minimum remain different claims.
2. Test selected lengths, bond dimensions, and stripe wavelengths. The $L = 96$ and 128 seeds are prepared, but no result from them is used here. Their planned fixed- $L = 64$ binding denominator isolates one finite-size question; it is not a new measurement of the larger ladder's binding ([Evidence 19](#)).
3. Release the separate ladder-density constraints and test transverse charge periods compatible with both charge and spin textures, as discussed in [Section 3.18](#). The completed A/B runs do not supply this control.
4. Check sensitivity to the mean-field range cutoff, currently $r_{\text{range}} = 4$ in the representative comparisons, and to weaker transverse hopping. These are model/approximation controls, separate from inner-DMRG convergence.
5. Complete amplitude and basis checks for the response pilot and measure response about a self-consistent normal reference at fixed density.
6. Separate connected fluctuations from disconnected order contributions. Subtracting $\langle O_i \rangle \langle O_j \rangle$ removes the elastic mean-field order contribution; both parts are needed to interpret structure factors. Extend the completed bond-class and reference-position comparisons to matched larger lengths and bond dimensions; the current effective slopes have substantial window dependence.

The completed square and unfrustrated-cubic grids now supply contrasting finite-run phase patterns. The finer cuts identify competing textures but do not resolve the order of the transition. The final September 19 review contains 38 completed 60-step run histories, or 2280 cell updates and 2400 individual ladder solves, all unaccepted; the earlier coarse square grid is a separate data set. No universal effect of V , certified ground-state phase diagram, transition temperature, phase stiffness, pair-density-wave state, or dynamical magnetic spectrum is claimed. The present contribution is a developing account of how local pairing tendencies and competing static orders relate in a specified coupled-ladder construction.

A Local evidence and drafting notes

A.1 Sources for project-specific statements

The following sources are repository documents and analysis artifacts, not external publications. Paths are relative to `ladder_mps_mft/docs`. The linked reports retain the underlying histories, definitions, and provenance checks. Their dated conclusions must be read together: later history analyses supersede the early suggestion of a unique paired $V = 0$ basin.

1. **Model, fields, and energy.** [Living methods notes](#), [variational functional](#), and [architecture](#). `METHODS_NOTES.tex`; `VARIATIONAL_FUNCTIONAL.md`; `ARCHITECTURE.md`.

2. **Literature and equation-level geometry correspondence.** [Literature review](#) and [source notes](#), including the September 8 comparison with the 2023 benchmark. `literature/literature_review.tex`; `literature/SOURCE_NOTES.md`.
3. **Isolated-ladder backbone and covariance survey, September 2.** [Stage 1 report](#). `reports/bare_stage1_t014_v0_20260902/REPORT.md`.
4. **Projected response discovery, September 3.** [Stage 2 report](#), with eigenvalues and reference-field summaries. `reports/bare_stage2_t014_v0_20260903/`.
5. **Cross-project scientific review, September 4.** [Review](#). Source for the response limitations, charge-gap correction, and proposed approximation controls; its campaign statuses are historical. `reports/systematic_review_20260904/REVIEW.md`.
6. **Bond-dimension comparison, September 5–6.** [Analysis](#), including the explicitly qualified September 6 energetic interpretation. `reports/chi400_comparison_20260905/ANALYSIS.md`.
7. **History-based revision of basin interpretation, September 8.** [Basin assessment](#). `reports/square_grid_20260908/BASIN_ASSESSMENT.md`.
8. **Square coverage and the flagged $(1.0, -0.4)$ endpoint, September 8.** [Grid description](#) and [divergence analysis](#). `reports/square_grid_20260908/`.
9. **Attractive two-family anchors, September 10.** [Analysis](#); complementary retrospective gate qualification in the [remainder contract](#). `reports/two_basin_vm04_20260910/README.md`.
10. **Zero-attraction anchors and slow texture relaxation, September 12–13.** [Analysis and follow-up](#). Figure 1 is reused without modifying the underlying data. `reports/two_basin_v000_20260912/README.md`.
11. **Combined synchronized campaign review, updated September 19.** [Combined report](#) and [complete cubic grid](#). Source inventories, physical profiles, gate checks and accounting coverage are retained with the detailed reports. The side-by-side figures use [matched physical histories](#) and their [source inventory](#).
12. **Complete coarse square grid and continuations.** [Updated square report](#), including both $(1.4, 0)$ continuations and the historical parent records.
13. **Finer square cuts and full energy curves, September 18.** [Cut report](#), [endpoint energy table](#), and [energy audit](#). Twenty endpoint sources include the twelve fine runs and eight coarse anchors. The [order-parameter audit](#) uses the same sources and consistent physical RMS windows.
14. **Square repulsive-interaction test, completed September 19.** [Four-seed comparison](#); all four spatial endpoints, with spin-node and energy diagnostics.
15. **All four trellis endpoints and alternating relaxation, September 19.** [Trellis report](#) and [spatial-cell method contract](#).
16. **Complete retrospective pair correlations, September 22.** [Correlation report](#), with full/connected and signed channel figures, source hashes, complete coverage, reference-window fits, bulk-cut sensitivity and stripe-wall metrics. `reports/pair_correlations_20260922/`.

17. **All four accepted square A/B results, September 22.** The same report’s completed square section and [gate/energy audit](#) supersede the September 21 first-result-only status. The original [preparation contract](#) retains the model, seeds and spatial-cell definition.
18. **Separate stripe-state and paired-state correlation grids, September 23.** [Definitions and analysis](#), [all 24 paired points](#), [all 42 stripe points](#), and [source/partition audit](#). All four manuscript grids reuse the completed measurements and retain the 42-state stripe group and 22 square states from the 24-state paired group; the two paired trellis entries remain in the source tables. No new solve or acceptance reclassification is implied.
19. **Prepared finite-size seeds, September 6.** [Seed description](#). Preparation evidence only; no larger-length simulation result is inferred. `reports/finite_size_seeds_20260906/REA DME.md`.

A.2 Reference provenance and manuscript development

The draft reuses stable keys from `../literature/references.bib`. The initial five additional sources in `additional_references.bib` are Tranquada et al. (2004), Vojta and Ulbricht (2004), Li et al. (2007), Berg et al. (2007), and Missiaen et al. (2025). Their primary arXiv and publisher records were checked on 15 September 2026. These additions support the discussion of magnetic spectra, planar coherence, and the stripe–pseudogap connection. Eleven further sources, checked on 18 September 2026, support the ladder-material, doping, optical-parameter, charge-order, NMR, neutron, and RIXS discussion in Section 2.4; reading depths and primary links are recorded in `SOURCE_NOTES.md`. Five additional numerical references, checked on 23 September, support the focused density-constraint discussion in Section 3.18; the source notes distinguish full-text checks from abstract-level background. The existing review’s reading-depth limits continue to apply to inherited references. On 24 September, the eleven material records were moved into the shared bibliography without changing their keys; thirteen further records and annotations broaden the material, exchange, pressure, structural, and trellis coverage. The Notbohm full-text parameters and the uniaxial onset criteria were checked explicitly. The stable `Scheie2025` key now cites its 2026 PRB publication. The review aims for broad coverage of the relevant material families, not a systematic database census or a new novelty assessment.

The introduction can be shortened for a target journal after the principal result and convergence controls are settled. The current results collection should retain its date and evidence qualifications until replaced by converged comparisons. A future manuscript should decide whether its central claim is interaction-driven order selection, geometry dependence, or the relation between a dressed normal-state response and the nonlinear ordered solutions. Those are candidate emphases, not additional results.

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